

A Detailed Investigation of the Disposition Effect and Individual Trading Behavior: A Panel Survival Approach

INGMAR NOLTE^a

^a*Financial Econometric Research Centre, Finance Group,
Warwick Business School, University of Warwick, UK*

Abstract

This paper uses a panel survival approach to analyze the trading behavior of foreign exchange traders. We concentrate on a detailed characterization of the shape of the disposition effect over the entire profit and loss regions. In doing so, we investigate the influence of a number of trading characteristics on the impact of the disposition effect. These trading characteristics include: special limit order strategies, trading success, size and the experience of our investors.

Our main findings are that i) the disposition effect has a non-linear shape. For small profits and losses we find an inverted disposition effect while for larger ones the usual positive disposition effect emerges. ii) The inverted disposition effect is driven to a great extent by patient and cautious investors closing their positions with special limit orders (take-profit and stop-loss). The normal positive disposition effect is found to be intensified for impatient investors closing their positions actively with market orders. iii) We show that unsuccessful investors reveal a stronger inverse disposition effect. iv) Evidence that bigger investors are less prone to the disposition effect than smaller investors is also found.

JEL classification: G10, C41, C32

Keywords: Panel Survival Models, Trading Activity Dataset, Foreign Exchange Market, Behavioral Finance, Disposition Effect

1 Introduction

In this paper we investigate investment decisions (open and close) of individual investors. The main focus of the paper is the characterization of the shape of the disposition effect over the complete profit and loss region. Our analysis uses detailed transaction data obtained from an electronic trading platform for currencies and we propose to model the investment decision process with a dynamic panel survival model.

A key feature of our model is the definition of investor's roundtrip and inactivity durations. Roundtrip durations are defined as the elapsed time from an initial entry into an investment to its complete exit. This means that we explicitly allow the trading strategy to be complex (e.g. several openings and/or several partial closings) and we do not restrict our attention to simple strategies (open followed by a full close) a priori. Inactivity durations are defined as the elapsed time between two roundtrips during which the investor has no exposure to a currency position.

Through our definition of roundtrip durations we address the problem of how to handle several successive openings and partial closings of an individual investor in a single security. This constitutes one of the key difficulties faced by studies that track individual investors' trading decisions on a high frequency level¹. The prevalent approach in the literature is to define durations between openings and closings by trying to match every single opening with its corresponding closing. Complications already arise in the case of matching partial closings. Moreover, such a matching procedure automatically introduces overlaps and thus dependencies that would require profound statistical techniques for inference and goodness-of-fit assessments.

Our definition of roundtrip durations is an elegant way to cope with this problem. Within the proposed panel survival model differences in effects for roundtrips relying on simple and complex trading strategies can easily be account for.

Beside this straightforward technical argument for our definition of roundtrip durations a mental accounting and narrow framing argument (Thaler (1985)) implicitly or explicitly invoked in many studies on individual trading behavior and behavioral biases (in particular the disposition effect) can be related to our setting. It is commonly assumed that especially individual investors consider their investment positions in an isolated form and do not take an overall portfolio view when making an investment decision.² In this paper we also follow this point of view and consider therefore periods with exposure (roundtrips) and periods without exposure to a currency position within a dynamic setup. We do, however, believe that the narrow framing argument should not be over-stretched and cannot be used to justify the consideration of *every* single transaction in a single currency pair in isolation. Such an approach ignores problems of overlaps, dynamics spillovers and difficulties of dealing with partial closings.

A main focus of this paper is the investigation of individual trading behavior and especially the disposition effect. The disposition effect (Shefrin and Statman (1985)) is considered to be the tendency that investors hold losing asset positions longer than corresponding

winning asset positions. There is an ongoing debate in the literature whether the disposition effect can be explained within a rational investor framework, with explanations such as informed trading, rebalancing or transaction costs put forward, or whether one should resort to behavioral theories such as the prospect theory³ of Kahneman and Tversky (1979).

Odean (1998a) shows that the most apparent approaches within a rational investor world seem to be unlikely explanations of the disposition effect. He provides evidence that his traders behave irrationally by selling winning positions too early and holding losing positions too long which in his analysis implies a real effect on investors' wealth. Locke and Mann (2005) also find that losing positions are held longer than winning positions, but in contrast to Odean (1998a) they cannot provide conclusive evidence that this trading behavior has significant negative impacts on investors' wealth.

Barberis and Xiong (2009) show that prospect theory preferences can predict a disposition in a rigorous theoretical model framework. They provide evidence in form of two simulation studies in which preferences are firstly defined over annual gains and losses, and secondly defined over realized gains and losses. In their studies it also turns out that for specific parameter constellations for expected returns and numbers of trading periods within a year an inverse disposition effect can be observed. Their evidence is based on the comparison of Odean (1998a)'s "proportion of gains realized (PGR)" and "proportion of losses realized (PLR)" measures with allow for an assessment of the mean disposition effect only.

Indeed, many of the existing studies use aggregated data to analyze the disposition effect for the mean (representative) investor applying a number of different methodologies. The early studies on investor behavior Lease et al. (1974), Schlarbaum et al. (1978a,b), and Shefrin and Statman (1985) as well as the studies of Badrinath and Lewellen (1991), Shapira and Venezia (2001) and Haigh and List (2005), compare mean durations of profitable and non-profitable investments. The studies of Odean (1998a) and Dhar and Zhu (2006) consider the difference between PGR and PLR measures within a portfolio setting. Grinblatt and Keloharju (2001) apply Logit models to estimate the probabilities of closing positions conditional on whether the position is traded either in profits or in losses.

The analyses on the disposition effect can furthermore be categorized into studies focusing on individual investors' trading behavior in stock markets (Lease et al. (1974), Schlarbaum et al. (1978a,b), Badrinath and Lewellen (1991), Odean (1998a), Chen et al. (2007), Feng and Seasholes (2005) and Ivkovich et al. (2005)), on the difference between individual and professional or institutional investors in stock markets (Locke and Mann (2005), Grinblatt and Keloharju (2001), Shapira and Venezia (2001), Dhar and Zhu (2006), Brown et al. (2006) and O'Connell and Teo (2009)), on investors in options and futures markets (Heisler (1994) and Heath et al. (1999), Locke and Onayev (2005)), on investors in the housing market (Genesove and Mayer (2001)), and on investors in experimental trading setups (Weber and Camerer (1998), Haigh and List (2005)). The studies of Kahneman (1992), Odean (1998a), Weber and Camerer (1998) and Barberis and Thaler (2003) discuss the choice of the correct reference point.

The modeling approach pursued in this paper can be seen in the context of survival or holding time approaches⁴. Besides the novel classification of roundtrip and inactivity periods, our approach contributes to this literature by using a sophisticated dynamic intensity model, that accounts for i) the panel characteristics of the data, ii) unobserved heterogeneity and iii) spill-over effects from inactivity to roundtrip periods and vice versa. Our approach allows for iv) inference on both the inactivity and the roundtrip states and v) the investigation of specific non-linearities (and factors driving these) in the disposition effect over the complete profit and loss region. Last but not least we show that our model is able to explain the underlying data generating process very well by a rigorous econometric assessment of its goodness-of-fit, which strengthens the reliability and the validity of our results.

The data used in this paper is unique and consists of the trading activity record of OANDA FXTrade, which is an internet trading platform for currencies. The set of investors on this platform is quite heterogeneous and ranges from retail up to smaller professional investors. From the trading activity record we construct a dataset of roundtrip and inactivity durations for all active investors during the period from 1st of October 2003 to 14th of May 2004 (227 days). The analysis presented in this paper focuses on the most actively traded currency pair – EUR/USD – which accounts for 61.5% of the total turnover on OANDA FXTrade during our sample period.

Our dataset is perfectly suited for analyzing individual investors' trading behavior, since in contrast to discount broker datasets, the number of trading instruments is strictly limited. Therefore, we can trace investors' trading strategies separately for every trading instrument (currency pair). By considering these trading strategies we do not need to account for different risk classes of the underlying instruments. Moreover, since we consider trading in the foreign exchange market, overnight effects are less severe and influences of short sale restrictions do not exist.

Although the trading proportion of OANDA FXTrade is relatively small in comparison to the overall FX market the results of this study do provide some interesting findings which possess relevance in a broader finance context. It worth mentioning here that the electronic retail trading segment is becoming more and more important. With advances in computer and internet technologies retail traders, smaller professional investors and institutions have nowadays easy instant access to all sorts of financial instruments. For retail investors it has been almost impossible to trade currencies or derivatives 20 years ago, without contacting their house bank. The understanding of the trading behavior of retail investors is of utmost importance for banks and markets makers designing and setting up electronic trading platforms. They face the challenge of hedging counterparty risk in the sense of whose investor's position will be kept on the bank's own inventory books and whose position should be hedged away. The design of this hedging algorithm on the basis of individual investors' characteristics and trading behavior is the "holy grail" of every electronic trading platform acting as a market maker.

Moreover, trading activity records such as the OANDA FXTrade one will become more and more available. They allow to trace all transactions of individual investors over time, thus embody the most valuable form of market microstructure information, and hence

provide deeper and more detailed insights into complex trading behavior on the micro level. The development of financial econometric approaches to handle this vast amount of information is still in its infancy.

The paper is organized as follows: Section 2 describes the construction of our dataset and provides a brief introduction into the trading mechanism on OANDA FXTrade. Section 3, presents the panel survival model. Section 4 contains the empirical analysis with a detailed interpretation of the results in the light of the disposition effect and further behavioral biases. Section 5 concludes.

2 Description of the Dataset

The dataset used in our analysis is constructed from the trading activity record of OANDA FXTrade, which ranges from 1st of October 2003 to 14th of May 2004 (227 days). Therein, all trading activities of 4893 different traders in 30 possible currency pairs is recorded on a second by second basis⁵. OANDA FXTrade is a 24/7 electronic trading platform mainly for retail and smaller professional investors and institutions. OANDA itself acts as a market maker and sets bid and ask quotes based on an external datafeed (crossing network) and a proprietary forecasting algorithm. A trader can submit market orders, limit orders and special limit orders (stop-loss, take profit) to the system. Limit orders are maintained for up to one month, and a trader may change submitted limits at any time without incurring an extra fee. Depending on the transaction type, we receive information on the transaction prices (market orders, limit orders executed, stop-loss, take profit, margin call), on the bid and ask quotes (pending limit orders), on the transaction volume and on the limits of stop-loss and take-profit orders. Moreover, we have information on each traders' invested start position per currency pair on 1st of October 2003 00:00:00 (Eastern Standard Time). For each trader we compute separately the respective currency pair durations from entering into a position until leaving it, i.e., the time during which he had an exposure to a specific currency pair position. We refer henceforth to this duration as the roundtrip (state) duration. Furthermore, we calculate the durations for periods in which the investor has not been invested in the specific currency pair to which we refer as the inactivity (state) duration. Thus, we obtain an alternating series of roundtrip and inactivity durations. Figure 1 clarifies the panel structure of the dataset constructed. We concentrate only on the most actively traded currency pair: EUR/USD which accounts for nearly 2/3 of all trading activity on OANDA FXTrade.

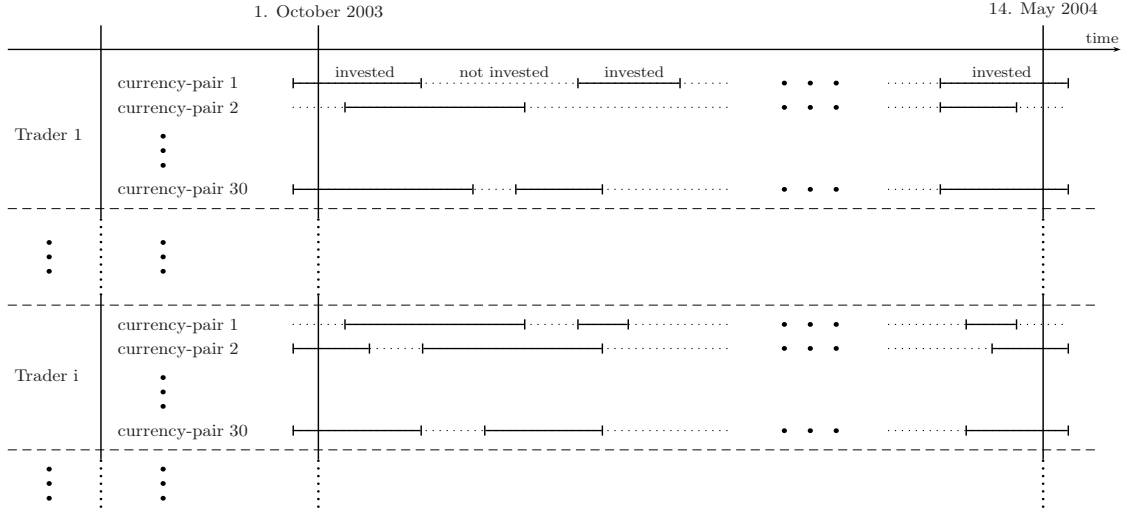


Figure 1: Structure of the panel dataset. The figure illustrates the alternating pattern of roundtrip (invested) and inactivity (not invested) durations.

3 The Model

We propose a panel survival model with two states: $m = R$ for roundtrips and $m = I$ for inactivity periods. Let $i, i = 1, \dots, N$ denote the i^{th} trader. For the i^{th} trader we observe C_i duration cycles, which are either roundtrip or inactivity durations. Let $c, c = 1, \dots, C_i$ denote the c^{th} cycle of the i^{th} trader. We do not need to distinguish between the origin and the destination state due to the alternating behavior of roundtrip and inactivity states. Thus, transitions from roundtrip to roundtrip as well as from inactivity to inactivity are excluded. We denote the indicator function by d_{ic}^m , which takes the value 1 if the i^{th} trader is in state m in the c^{th} cycle and 0 otherwise. Let τ_c^i denote the duration of the c^{th} cycle of the i^{th} trader and let t_c^i denote the calendar time at which the i^{th} trader leaves the c^{th} cycle. Let

$$f(\tau_c^i | \mathfrak{F}_{t_c^i}) \quad (1)$$

denote the conditional density of the duration τ_c^i in which we condition on information represented by $\mathfrak{F}_t$ being available at the end of the respective duration at time t_c^i . For the ease of notation we suppress the particular conditioning information and consider $f(\tau_c^i) \equiv f(\tau_c^i | \mathfrak{F}_{t_c^i})$ in the following. We can then decompose $f(\tau_c^i)$ into

$$f(\tau_c^i) = \prod_{m \in \{I, R\}} f^m(\tau_c^i)^{d_{ic}^m}, \quad (2)$$

where $f^m(\tau_c^i)$ denotes the (conditional) m^{th} state duration density, which can be further decomposed into

$$f^m(\tau_c^i) = \theta_{ic}^m(t_c^i, \tau_c^i) \bar{F}^m(t_c^i, \tau_c^i). \quad (3)$$

Here $\theta_{ic}^m(t_c^i, \tau_c^i)$, denotes the (conditional) intensity, i.e. the instantaneous probability at time t_c^i , of the i^{th} trader for leaving the m^{th} state in the c^{th} cycle and $\bar{F}^m(t_c^i, \tau_c^i)$ the corresponding (conditional) m^{th} state survivor function, i.e. the probability that the i^{th}

trader has not left the m^{th} state in the c^{th} cycle at time t_c^i and thus survived in that state from $t_c^i - \tau_c^i$ to t_c^i . Hence $f(\tau_c^i)$ becomes

$$f(\tau_c^i) = \prod_{m \in \{I, R\}} (\theta_{ic}^m(t_c^i, \tau_c^i) \bar{F}^m(t_c^i, \tau_c^i))^{d_{ic}^m}. \quad (4)$$

$\bar{F}^m(t_c^i, \tau_c^i)$ can be expressed in terms of the corresponding integrated intensity given by

$$z_{ic}^m(t_c^i, \tau_c^i) = \int_0^{\tau_c^i} \theta_{ic}^m(t_c^i, \nu) d\nu \quad (5)$$

as

$$\bar{F}^m(t_c^i, \tau_c^i) = \exp(-z_{ic}^m(t_c^i, \tau_c^i)), \quad (6)$$

so that the complete log-likelihood of the model is given by

$$\ln \mathcal{L}(W, \beta) = \sum_{i=1}^N \sum_{c=1}^{C_i} \sum_{m \in \{I, R\}} d_{ic}^m (\ln \theta_{ic}^m(t_c^i, \tau_c^i) - z_{ic}^m(t_c^i, \tau_c^i)), \quad (7)$$

where W generically refers to the data and β is the generic symbol for all relevant parameters. We propose a parameterization of $\theta_{ic}^m(t_c^i, \tau_c^i)$ consisting of a Burr intensity component being multiplicatively linked to an exponentially transformed flexible Fourier form (FFF) (Gallant (1981)) in the following way:

$$\theta_{ic}^m(t_c^i, \tau_c^i) \equiv \frac{\exp(\beta^m x_{ic}^m) \alpha^m (\tau_c^i)^{\alpha^m - 1}}{1 + \sigma^{m2} \exp(\beta^m x_{ic}^m) (\tau_c^i)^{\alpha^m}} \cdot \exp \left(\sum_{k=1}^K \nu_{k,s}^m \sin(2k\pi u(\tau_c^i)) + \nu_{k,c}^m \cos(2k\pi u(\tau_c^i)) \right) \quad \text{for } m = 1, 2, \quad (8)$$

where x_{ic}^m denotes the vector of state specific explanatory variables of the i^{th} trader in the c^{th} cycle, and β^m denotes the corresponding (state dependent) parameter vector. The variables employed in the empirical application are detailed in Table 1. α^m and σ^{m2} denote the shape and scaling parameter of the Burr distribution. K denotes the order of the exponential FFF. $\nu_{k,\cdot}^m$ represent the coefficients associated with the sine and cosine terms, and $u(s) \equiv 1 - \exp(-s)$ is the transformation of s (duration or the backward recurrence time) with the property that $u(s) \in [0, 1]$.

The proposed specification is quite flexible in capturing the shape of the baseline intensity. The model nests the simple Burr specification if all $\nu_{k,\cdot}^m$ coefficients are equal to zero, the simple Weibull model if $\sigma^{m2} = 0$, the simple Log-Logistic model if additionally $\sigma^{m2} = 1$ and the simple exponential model if $\sigma^{m2} = 0$ and $\alpha^m = 1$ as well as their corresponding exponential FFF extensions for the case where $\nu_{k,\cdot}^m$ are not equal to zero but the associated parameter restrictions on σ^{m2} and α^m are fulfilled. The simple Burr specification can be interpreted as mixed proportional intensity model in which the Weibull density is multiplicatively mixed with a Gamma distributed random variable v with mean 1 and

variance σ^{m2} . The variable v can be interpreted as a random effect capturing unobserved heterogeneity among our investors. Thus, a test of the Burr against the Weibull model can be interpreted as a test for unobserved heterogeneity. Failing to account for unobserved heterogeneity usually results in inconsistent, biased and inefficient parameter estimates.

The desired flexibility of the intensity specification comes at the cost of Quasi Maximum Likelihood failing to justify parameter consistency and we have to assume that the proposed specification captures the true data generating process so that we can resort to Maximum Likelihood reasoning for parameter consistency. To this end, under the assumption of the correct specification for θ_{ic}^m , the corresponding integrated intensities z_{ic}^m are unit exponentially distributed and we consider $\hat{z}_{ic}^m$ as generalized model residuals which asymptotically should also be unit exponentially distributed if the model is correctly specified allowing for inference as to whether the proposed intensity specification is correct.

4 Empirical Analysis

To address the heterogeneity of our traders we collect them together according to their total transaction volume over the whole period into 20 groups. The first group contains traders with the smallest transaction volume (0% - 5% of the transaction volume distribution), and the last group (95% - 100%) the traders with the largest total volume. This categorization eases the estimation of the proposed intensity based model and allows us to draw conclusions on the differing trading behavior across groups.

Descriptive Statistics

The descriptive statistics for all 20 investor groups are presented in Table 4 in the Appendix. Descriptive statistics on the roundtrip and the inactivity state durations, the durations separately for loss-making and profit-making roundtrips, the profit and loss, the absolute profit and loss and the transaction volume are reported. Furthermore, we construct dummy variables and report the percentages of whether a roundtrip is closed with a special limit order (stop-loss and take profit), whether the roundtrip involved a complex trading strategy and whether the investor has been profitable (successful) over each of the two past roundtrips. Under a complex trading strategy we understand any trading pattern consisting not just of an initial open followed by a full close.

From the 20 ordered investor groups we observe that the mean transaction volume involved in a roundtrip ranges from 144 EUR for the smallest group to around 370000 EUR for the largest groups. The maximum transaction volume ranges from 4800 EUR to 53 Mil. EUR. These figures highlight the heterogenous composition of the investors on OANDA FXTrade with small retail and private investors up to professional traders. Across all investor groups we observe that a mean roundtrip duration is roughly half the length of an inactivity period, and that from group 1 to 20 the length of a mean roundtrip duration steadily decreases from 30 to 7 hours. Thus, the small investors do not only trade with smaller volumes they also trade less frequently than the bigger investors. Interestingly, the small investors also trade more complicated strategies than the bigger investors as indicated by the percentages of complex roundtrips, which decline from 50% for group

1 to 25% for group 20. Across all investor groups we do not observe big differences in the percentages of roundtrips closed with special limit orders (stop-loss or take-profit) – around 20% – and differences in the percentages of successful trading which lie around 45%.

A simple comparison of the mean lengths of loss-making roundtrips with those of profit-making roundtrips suggests a disposition effect for all investor groups, except groups 8 and 14, since the means of the loss-making roundtrips are longer than the means of the profit-making roundtrips. However, comparing the corresponding median values often reveals (except for groups 9, 18, 19, 20) an inverse disposition effect since the median loss-making roundtrips are shorter than the median profit-making ones. A more detailed look at the higher quantiles shows for all groups an inverse disposition effect up to a certain quantile (for groups 2, 7, 8, 10, 14, 15 even up to the 95% quantile) and a positive disposition effect from that quantile onwards.

These observations indicate, from a purely descriptive point of view, that mean duration comparisons such as those typically carried out in empirical studies which investigate the disposition effect can be misleading and might yield a distorted and incomplete picture of the true form of the disposition effect. We observe a very clear non-linearity in the disposition effect over the profit and loss region, which will be further investigated in the proposed intensity framework. Here, we only compared the distributions of the lengths of the roundtrips if the roundtrip has been profitable or not. The estimates of the intensity model in contrast will enable us to compare lengths of profitable and loss-making roundtrips at predetermined profit and loss values ± 1000 EUR, say, which will yield a much more detailed picture of the non-linearity of the disposition effect over the entire profit and loss region, while controlling for the influence and interaction of further explanatory variables describing trading characteristics.

Estimation Results

The detailed maximum likelihood estimation results of the panel intensity model for all 20 investor groups are presented in Table 5 in the Appendix. In the following we only present statistics and figures for investor groups 1, 7, 14 & 20⁶. We include the explanatory variables x_{ic}^m described in Table 1 for the roundtrip ($m = R$) and the inactivity state ($m = I$) into the model setup presented in equations (7) and (8).

Variable Description	Roundtrip Par.	Inactivity Par.
constant	c^R	c^I
dummy – if successfull	c_{sc}^R	c_{sc}^I
duration of previous roundtrip	$\beta_{dr(-1)}^R$	$\beta_{dr(-1)}^I$
duration of previous inactivity period	$\beta_{di(-1)}^R$	$\beta_{di(-1)}^I$
maximum invested volume during roundtrip	β_{vol}^R	–
dummy – if special limit order close	c_{sl}^R	–
dummy – if complex trading strategy	c_{cx}^R	–
profit over roundtrip	β_p^R	–
profit ² over roundtrip	$\beta_{p^2}^R$	–
profit ³ over roundtrip	$\beta_{p^3}^R$	–
interaction – profit & successfull	β_{p*sc}^R	–
interaction – profit & special limit order close	β_{p*sl}^R	–
interaction – profit & complex trading strategy	β_{p*cx}^R	–
loss over roundtrip	β_l^R	–
loss ² over roundtrip	$\beta_{l^2}^R$	–
loss ³ over roundtrip	$\beta_{l^3}^R$	–
interaction – loss & successfull	β_{l*sc}^R	–
interaction – loss & special limit order close	β_{l*sl}^R	–
interaction – loss & complex trading strategy	β_{l*cx}^R	–

Table 1: Description of explanatory variables and associated parameters. All explanatory variables except the constants are trader i and cycle c specific.

The set of variables included in the model is motivated the descriptive analysis and intends to capture investor trading characteristics and potential non-linearities in the shape of the disposition effect over the complete profit and loss region. Before we continue with the interpretation of the estimation results we need to consider the goodness-of-fit of the estimated model. We clearly observe for all 20 groups that the parameters $\nu_{k.}^m$ of the exponential FFF are jointly highly significant for both the roundtrip and inactivity state. The null hypothesis that the scaling parameter $\sigma^{m2} = 0$, which is the test of a Weibull against a Burr specification cannot be rejected for 2 groups only on the 10% level for the roundtrip component and for 8 groups for the inactivity component. Thus, it seems that a Weibull specification alone is not flexible enough to explain the baseline intensity component, which indicates that there is a considerable part of unobserved heterogeneity which is captured by the mixture of the Weibull density with a Gamma distributed random variable with unit mean and variance σ^{m2} . Unobserved heterogeneity arises because the explanatory variables included into the model are not capable of fully explaining the existing heterogeneity prevalent in the trading behavior of our investors. This is actually not surprising since the groups of investors considered in this study range from small retail up to professional investors. Even within the groups it turns out that the investors are quite different so that an additional stochastic factor trying to capture these differences seems appropriate. Unobserved heterogeneity can therefore to a large extent be associated with an omitted variable problem and failing to account for it usually renders estimation results and model inference invalid.

The estimated baseline intensities (groups 1, 7, 14 & 20) for the roundtrip states are depicted in Figure 2. We observe a very similar pattern across all investor groups: a declining

baseline intensity function, with local minima after half a day and one-and-a-half days and local maxima after one day and two days. Intuitively the baseline intensity describes how frequently we observe another opening or closing decision (here in alternating form) for the investor under consideration. Since the pattern depicted in the graphs is very regular with maxima at multiples of full days we see that it reflects habit persistency in trading behavior and preferences over trading times, which might in part be explained by work schedules, regular office hours and overnight effects.

In Table 2 we report the means and the standard deviations of the raw roundtrip and inactivity duration series (pooled over investors) and their associated residual series. Figures 3 and 4 depict the corresponding QQ-plots for the roundtrip and inactivity residual series for groups 1, 7, 14 & 20. Although we observe a slight underdispersion in the inactivity state residual series, we see that the roundtrip state residual series are very close to a unit exponential distribution. Altogether, the properties of the residual series show an extremely good fit of the proposed Burr baseline intensity model extended by an exponential FFF.⁷

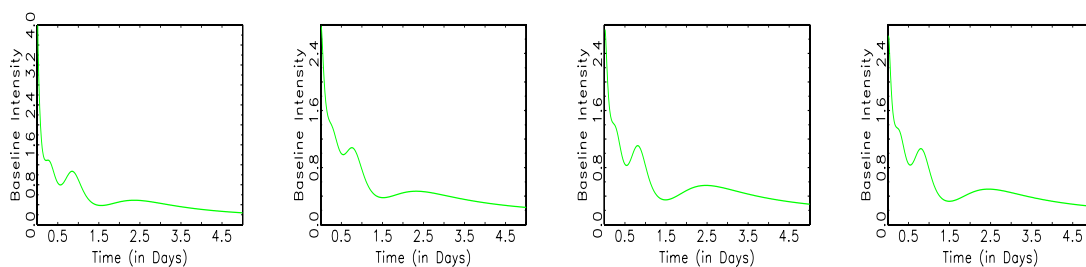


Figure 2: Estimated baseline intensities for investor groups (from left to right) 1, 7, 14 & 20.

	roundtrip		inactivity			roundtrip		inactivity	
	raw	res	raw	res		raw	res	raw	res
	1					7			
mean	1.29	1.00	2.46	1.00		0.46	1.00	0.83	1.00
std	4.62	0.98	9.00	0.93		2.65	0.98	4.42	0.95
	14					20			
mean	0.31	1.00	0.67	1.00		0.33	1.00	0.52	1.00
std	1.88	0.98	3.28	0.96		2.18	0.97	2.59	0.95

Table 2: Mean and standard deviation of the raw and the residual series for the EUR/USD currency pair of the roundtrip and inactivity states for investor groups 1, 7, 14 & 20.

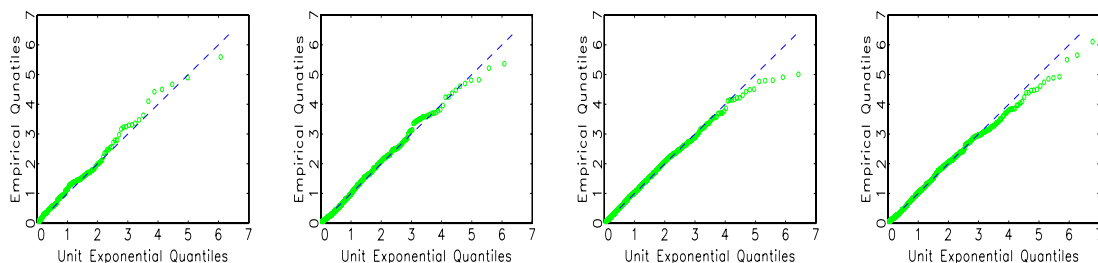


Figure 3: Quantile-Quantile plots of the roundtrip residual series against the unit exponential distribution for the EUR/USD currency pair. The plots correspond from left to right to investor groups 1, 7, 14 & 20.

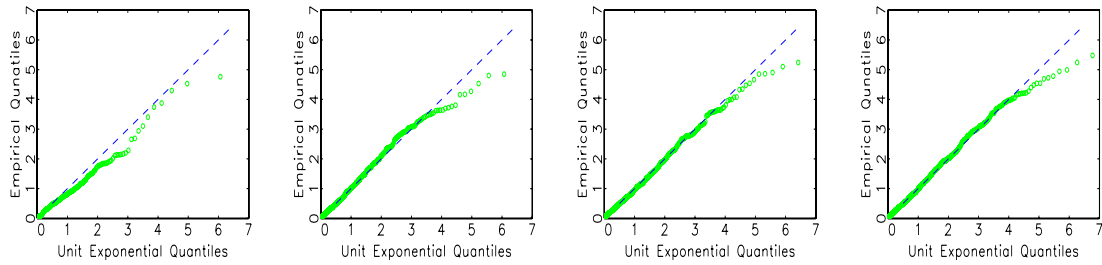


Figure 4: Quantile-Quantile plots of the inactivity residual series against the unit exponential distribution for the EUR/USD currency pair. The plots correspond from left to right to investor groups 1, 7, 14 & 20.

Interpretation of the Estimation Results and a Detailed Look at the Disposition Effect

The parameter estimates $\beta_{di(-1)}^R$, $\beta_{dr(-1)}^R$, $\beta_{di(-1)}^I$ and $\beta_{dr(-1)}^R$ (reported in Table 5 in the Appendix) that capture the influence of the previous roundtrip and inactivity durations on both the current roundtrip and inactivity intensity are for all 20 investor groups always highly (1% level) significantly negative. Thus, we observe a very clear clustering pattern in the trading activity since longer (shorter) durations are followed by lower (higher) intensities for exiting the roundtrip and the inactivity state. This observation clearly shows that there are successive periods in which investors actively trade more frequently and successive periods in which they don't. Such a dynamics pattern cannot simply be ignored and should be addressed in any study concerned with holding times or intensities especially for the disposition effect. Within our proposed modelling framework this is easily achieved.

c_{cx}^R which is the coefficient on the dummy variable, which takes on the value of one if a roundtrip represents a complex trading strategy, is also for all 20 investor groups highly significant and negative. Thus, this coefficient controls for the effect that roundtrips involving a more complex trading strategy are simply longer.

c_{sl}^R is the coefficient on the dummy variable indicating whether a roundtrip is closed by a special limit (stop-loss or take-profit) order. Special limit orders in contrast to standard limit orders are usually submitted on OANDA FXTrade jointly with the opening order for the position. Hence, they reflect extremely cautious and passive trading behavior, since the submitter wants to protect himself from severe losses or wants to realize a profit when a certain threshold is hit right from the very beginning. In both cases, it is less likely that the trader will actively monitor the market and immediately react to new information, than if he had not submitted the special limit order. On the one hand, such behavior can be interpreted either as quite sophisticated and experienced trading since the trader is aware of the fact that he will not be able to follow the market and thus ensures his positions against periods of high risk. On the other hand, it can be interpreted as uniformed and inexperienced trading since the trader does not expect to have access to private information which he could exploit with an active trading strategy. Nevertheless, when a roundtrip is closed by executing a special limit order the profit and loss region was bounded at least in one direction a priori. The sign of c_{sl}^R is always negative and significantly (5% level) negative for 18 investor groups. Thus, c_{sl}^R captures the tendency that roundtrips, which

are closed by special limit orders and rely in part on passive trading strategies are longer. This is another important trading characteristic that needs to be account for and whose cross-effects with other variables (especially profit or loss variables) should not be ignored.

c_{sc}^R is the coefficient on the dummy variable, that takes on the value one if the investor has been profitable (successful) in the last two roundtrips. On a 5% significance level the coefficient c_{sc}^R is significant only for 6 groups. However, whenever c_{sc}^R is significant it is positive, which shows that successful investors generate higher trading activity. Numerous theoretical models (for instance Daniel et al. (1998), Odean (1998b), Wang (1998), Gervais and Odean (2001) and Scheinkman and Xiong (2003)) put overconfidence forward as an explanation for excess trading volumes and empirical evidence is provided by De Bondt and Thaler (1995), Barber and Odean (2001) and Statman et al. (2006) with low frequency data. Self-attribution bias (Wolosin et al. (1973)) because of past trading success can be considered as one facet of overconfidence caused by an incorrect interpretation of past investment success. It is usually distinguished from overconfidence caused by biased perceptions of the precision of private information (Alpert and Raiffa (1982) and Lichtenstein et al. (1982)). Hence, the above finding can be seen as reflecting the effect of self-attribution bias on trading activity.

The coefficient β_{vol}^R which captures the effect of the maximum invested trading volume during a roundtrip is positive at a 1% significance level for all investor groups. It very clearly indicates that higher trading volumes imply shorter roundtrip durations, which reflects an increasing degree of overconfidence given by trading volume also implies more trading activity.

Let us now address the disposition effect directly. In terms of the roundtrip intensity we consider the following definition of the disposition effect.

Definition: Disposition Effect

The tendency to hold losses longer than corresponding profits, reformulated in terms of the roundtrip intensity, is that there is a smaller intensity (instantaneous probability to leave the roundtrip state) for losses than for corresponding profits.

This definition only requires that for a given loss there is a smaller intensity than for the corresponding profit. It does not require that the difference between the intensity for losses and the intensity for profits is either constant over the length of the holding period nor constant over the associated monetary amount. Absence of a disposition effect is hence described by a roundtrip intensity which is axially symmetric at zero over the profit and loss region. In this case the probability of leaving the roundtrip state would be exactly the same for a $x > 0$ units-of-the-base-currency-loss as for a $x > 0$ units-of-the-base-currency-profit.

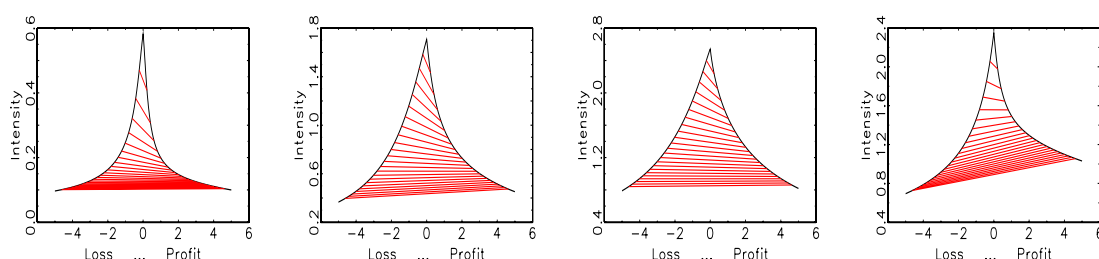


Figure 5: Illustration of the disposition effect for EUR/USD. The plots correspond, from left to right, to investor groups 1, 7, 14 & 20. Each plot shows the intensity to leave the roundtrip state (y-axis) over a standardized profit/loss region (x-axis). The points -1 and 1 on the x-axis correspond to the negative and positive mean absolute profit/loss, respectively, realized in the associated group. The upward (downward) sloping connecting lines indicate a positive (negative) disposition effect in this region.

In Figure 5 we show for investor groups 1, 7, 14 & 20 the estimated roundtrip intensity as a function of the profit and loss. All further explanatory variables have been set to their mean values. The graphs are standardized so that the points -1 and 1 on the x-axis correspond to the negative and positive mean absolute profit/loss, respectively, realized in the associated group⁸. The outer values ± 5 appear to roughly correspond to a quantile between the 95% and the 98% quantile of the absolute profit/loss distribution. The upward (downward) sloping lines in the inner region indicate a positive (negative) disposition effect with respect to that particular profit/loss size.

For all groups, we observe, an inverse disposition effect (downward sloping lines) for small profits and losses and especially for the smaller investor groups (1-8), the usual disposition effect for larger profits and losses (upward sloping lines). The bigger investor groups (12-19) seem not to reveal any disposition effect even for larger profits and losses and group 20 – the group with the biggest investors – again shows a disposition effect for larger profits and losses, which might be explained by several extremely high losses with long roundtrip durations within this group.

In general, we observe a highly non-linear shape to the disposition effect which is inverted for small profit and loss amounts and positive for larger ones. There, is also a tendency for the larger investors to show smaller disposition effects and the inverse relationship also holds.

In the literature several studies (c.f. Odean (1998a), Shapira and Venezia (2001) and Grinblatt and Keloharju (2001)) document that the “average investor” exhibits a disposition effect and the studies of Odean (1998a), Ivkovich et al. (2005), Feng and Seasholes (2005) stress an tax argument (e.g. Stiglitz (1983) and Constantinides (1984)) that the disposition effect is diminished, disappears, or is inverted, since in the presence of taxes losses should be immediately realized after they arise. However, Dhar and Zhu (2006) show that even in the absence of tax effects, up to one fifth of their investors exhibit an inverse disposition effect. Barberis and Xiong (2009) show in their simulation studies that even the mean investor can possess an inverse disposition effect for specific parameter constellations of the expected return and monitoring frequencies, but their model assumes full rationality.

Without stressing the tax argument we investigate a straightforward insurance argument as an alternative and/or complementary explanation for the existence of an inverted disposition effect: Investors might want to insure themselves against large losses by submitting limit orders with tight limits which in turn would imply a diminished or an inverse disposition effect for small losses. Stop-loss orders represent the most aggressive form of such limit orders and their effect on the disposition effect is analyzed below. We can compare profit-making roundtrips closed by take-profit orders with loss-making roundtrips closed with stop-loss orders. Osler (2003) also argues that the use of stop-loss orders can be completely rational in a trading scenario in which other traders already use them. Following this line of argument we should observe a diminished disposition effect and therefore less behavioral bias in this case.

A potential counter argument for this point of view is provided by adopting the usual convention of Glosten (1994) and Seppi (1997), that roundtrips closed by special limit orders can be considered as roundtrips being completed by (temporarily) patient or uninformed investors, which accordingly should be more prone to behavioral bias and thus reveal a higher degree of disposition. A large part of the literature on the disposition effect and further behavioral biases in general concentrate on the discrimination between the levels on which investors are affected by behavioral biases. Shapira and Venezia (2001) find that individual (less informed and less sophisticated) investors are more affected by the disposition effect than institutional investors supporting the statement of Ross (1999) that institutional investors are more rational. Dhar and Zhu (2006) find that wealthier, more experienced and professional investors exhibit a smaller disposition effect. Locke and Mann (2005), Brown et al. (2006) and O'Connell and Teo (2009), however, provide evidence that institutional and professional investors are subject to behavioral biases as well. A similar result is found for less sophisticated and experienced Chinese investors by Chen et al. (2007). They find that their investors are contemporaneously subject to several behavioral biases such as the disposition effect, overconfidence, representative biases and narrow framing. The results of our analysis discussed above also provide only mixed evidence for the tendency that more sophisticated and experienced investors, according to their transaction size, possess a smaller disposition effect. We investigate below whether unsuccessful investors possess a stronger form of the disposition effect, which in part would complement the finding that less sophisticated investors are more prone to a disposition effect.

The impact of special limit orders (stop-loss and take-profit) on the disposition effect is depicted in Figure 6. In this figure, we directly plot the difference between the roundtrip intensity for the profit region and the roundtrip intensity for the loss region again for all investor groups. Hence, a positive value describes a positive disposition effect and a negative value an inverse one. The solid line illustrates the shape of the disposition effect if roundtrips are closed by special limit orders and the short dashed line if not. Both lines are the result of the coefficients β_{p*sl}^R and β_{l*sl}^R of the corresponding interaction terms. The whiskers at the bottom of the plots represent the absolute profit and loss distribution. Again, the graphs are presented in a standardized way across all investor groups, the value 1 on the x-axis always represents the mean absolute profit/loss within a group and the value 5 roughly corresponds to a quantile between the 95% and 98% quantiles

of the absolute profit/loss distribution. We observe, very clearly and consistently across all investor groups that roundtrips, which are closed by special limit orders, show a less pronounced disposition effect, that is always negative for all investor groups. Thus, we find clear evidence that the general inverse disposition effect for small profit and losses observed above is driven by roundtrips closed with special limit orders, representing the trading behavior of cautious, patient and maybe uniformed investors relying on a passive trading strategy. On the contrary, the general positive disposition effect for larger profits and losses is more pronounced for roundtrips not closed by special limit orders and thus mainly driven by market orders being submitted by traders in periods in which they actively track the market. Moreover, we observe across all investor groups that the distance between the short-dashed and the solid line becomes larger, the larger the investor. This indicates a higher impact of the type of order being used to close a roundtrip on the disposition effect for larger than for smaller investors. This result is highly significant according to the Likelihood-Ratio (LR) tests presented in the first panel of Table 3, which tests whether the two lines are significantly different from each other.

The effect of past trading success on the disposition is presented in Figure 7. The construction of the graphs is the same as for special limit orders. The solid line represents the effect if the roundtrip is closed by an investor who has been profitable in both of the last two roundtrips and the short-dashed line if not. We observe for all investor groups, except groups 3,4 and 8, the tendency for a more pronounced negative disposition effect over the complete profit/loss region if the roundtrip is closed by an investor who has not been successful (short-dashed line). This tendency also seems to be increasing over the size of investors since the distance between the solid and the short-dashed lines becomes wider the larger the investors. It is further reflected by the LR-tests in the second panel of Table 3 in which the effect of past trading success on the disposition effect is for smaller investor groups not significant or can be rejected at low significance levels. Thus, we find that unsuccessful investors reveal a stronger inverse disposition effect.

A further interesting question which can be analyzed in our model setup is whether investors with complex trading strategies reveal a different disposition effect than investors trading with simple open-hold-full-close strategies. Figure 8 depicts the corresponding graphs concerning this effect. Although, there is no general effect of complex trading on the disposition effects, the graphs reveal a tendency that for small profit and losses the inverse disposition effect is intensified if roundtrips rely on a complex trading strategy. For larger profit and losses this effect is then often reversed strengthening a positive disposition effect in this region. The corresponding LR-tests reveal that this pattern is only significant at the 10% level for 10 out of the 20 groups. Thus, we find only a slight indication that investors who employ complex trading strategies possess a smaller disposition effect. This observation certainly needs deeper investigation, but is generally in line with the finding that more experienced investors, who might apply more complex trading algorithms, possess a less pronounced disposition effect and hence complements the finding on the general distribution of the disposition effect over investor groups and the results on overconfidence.

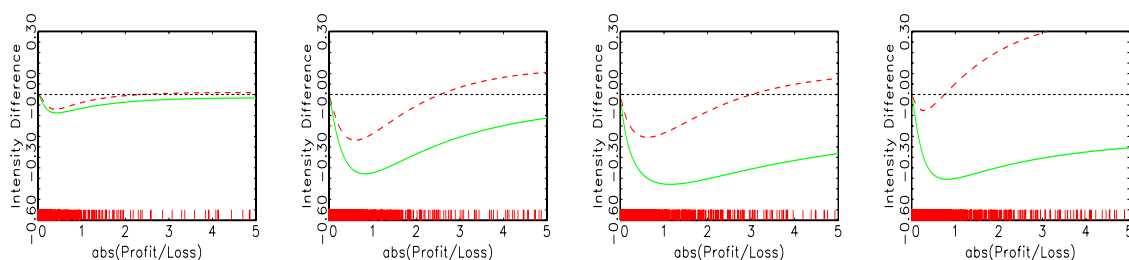


Figure 6: Impact of special limit orders on the disposition effect for the EUR/USD currency pair. The plots correspond from left to right to investor groups 1, 7, 14 & 20. Each plot shows the difference of the roundtrip intensity for profits and the roundtrip intensity for losses. The point 1 on the x-axis corresponds to the mean absolute profit/loss realized in the associated group. The solid line represents that a roundtrip is closed by a special limit order and the short dashed line if not. The whiskers at the bottom of the graphs represent the absolute profit/loss distribution.

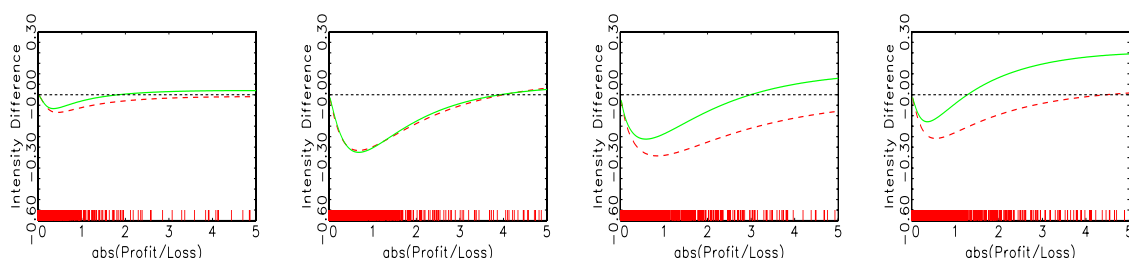


Figure 7: Impact of past trading success on the disposition effect for the EUR/USD currency pair. The plots correspond from left to right to investor groups 1, 7, 14 & 20. Each plot shows the difference of the roundtrip intensity for profits and the roundtrip intensity for losses. The point 1 on the x-axis corresponds to the mean absolute profit/loss realized in the associated group. The solid line represents that a roundtrip is closed by an overconfident investor and the short dashed line if not. The whiskers at the bottom of the graphs represent the absolute profit/loss distribution.

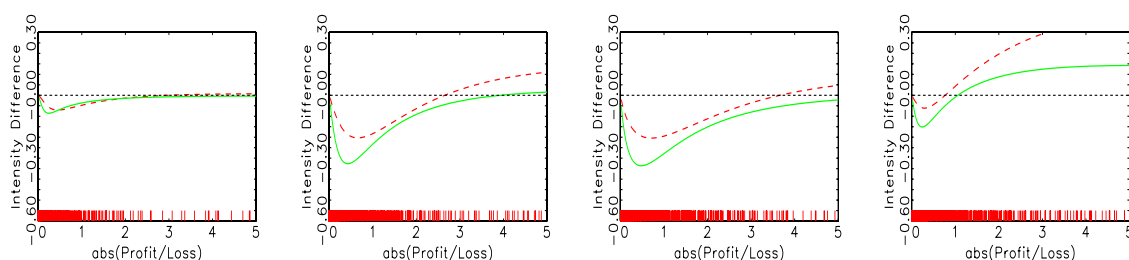


Figure 8: Impact of complex trading strategies during a roundtrip on the disposition effect for the EUR/USD currency pair. The plots correspond from left to right to investor groups 1, 7, 14 & 20. Each plot shows the difference of the roundtrip intensity for profits and the roundtrip intensity for losses. The point 1 on the x-axis corresponds to the mean absolute profit/loss realized in the associated group. The solid line represents that a complex trading strategy is applied and the short dashed line if not. The whiskers at the bottom of the graphs represent the absolute profit/loss distribution.

Likelihood Ratio Test: special limit order close										
	1	2	3	4	5	6	7	8	9	10
LR-Stat	93.6239	6.5816	20.8089	52.1358	49.5902	59.4559	28.9696	23.7222	10.4875	37.8483
p-value	0.0000	0.0372	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0053	0.0000
	11	12	13	14	15	16	17	18	19	20
LR-Stat	54.4719	44.8365	103.5667	95.0349	9.5744	174.2150	50.0627	46.3760	42.0606	81.4059
p-value	0.0000	0.0000	0.0000	0.0000	0.0083	0.0000	0.0000	0.0000	0.0000	0.0000
Likelihood Ratio Test: past trading success										
	1	2	3	4	5	6	7	8	9	10
LR-Stat	2.1367	2.9021	10.2760	1.3454	0.8577	8.6168	63.7331	14.7605	23.3056	2.3655
p-value	0.3436	0.2343	0.0059	0.5103	0.6513	0.0135	0.0000	0.0006	0.0000	0.3064
	11	12	13	14	15	16	17	18	19	20
LR-Stat	43.1236	23.9128	17.5150	5.1972	10.4448	15.9229	33.0603	21.0800	114.9656	14.8011
p-value	0.0000	0.0000	0.0002	0.0744	0.0054	0.0003	0.0000	0.0000	0.0000	0.0006
Likelihood Ratio Test: complex trading strategy										
	1	2	3	4	5	6	7	8	9	10
LR-Stat	3.2407	3.3686	5.3949	11.4362	0.7800	2.5850	1.5802	8.9617	23.8882	7.6879
p-value	0.1978	0.1856	0.0674	0.0033	0.9618	0.2746	0.4538	0.0113	0.0000	0.0214
	11	12	13	14	15	16	17	18	19	20
LR-Stat	0.1195	8.3695	7.6152	2.9698	60.0576	1.8733	4.7229	3.3728	21.4976	2.1144
p-value	0.9420	0.0152	0.0222	0.2265	0.0000	0.3919	0.0943	0.1852	0.0000	0.3474

Table 3: Test statistics and p-values for the likelihood ratio test for the full model specification against the restricted model specifications (special limit order close, past trading success and complex trading strategy) for EUR/USD. The test statistics for all tests are χ^2_2 -distributed.

5 Conclusion

This paper investigates the high-frequency trading behavior of a heterogeneous set of foreign exchange market traders on an electronic trading platform (OANDA FXTrade) by means of a panel survival approach.

Individual trading processes are decomposed into roundtrip periods defined as the time between entering into a position and leaving it, and inactivity periods are defined as the time between two roundtrips. A panel survival model relying on a flexible mixed proportional intensity specification accounting for individual unobserved heterogeneity is used to model the data generating process of these two duration processes. We show that our modeling framework is well suited to characterizing the properties of the data and that the proposed Burr intensity specification extended by an exponentially transformed flexible Fourier form describes the baseline intensity well.

Time is the central element in our model which we use to investigate individual trading behavior and the disposition effect. We have focussed on the detailed characterization of the shape of the disposition effect over the full profit and loss region, the links between the disposition effect and special limit orders, past trading success and complex trading behavior as well as assessing the disposition effect across the size and the degree of experience of our investors.

Our main findings are that

- i) the general disposition effect has a non-linear shape. For small profits and losses we find an inverted disposition effect and for larger profit and losses the usual positive disposition effect.
- ii) The inverted disposition effect is to a great extent driven by patient and cautious investors closing their positions with special limit orders (take-profit and stop-loss),

whereas the positive disposition effect is found to be intensified for impatient investors closing their positions actively with market orders.

- iii) In addition, we find that unsuccessful investors display a stronger inverted disposition effect.
- iv) Evidence that bigger investors are less prone to the disposition effect than smaller ones is provided and
- v) we find indication that investors relying on complex trading strategies are less affected by the disposition effect.

The approach presented in this paper can be extended in several dimensions: A multivariate specification in which the duration processes of several currency pairs are modelled jointly could provide a better characterization of the individual trading behavior and perhaps deeper insights into individual trading preferences and the motivation to open and close positions taking the cross effects with further currency pairs into account. Although the proposed intensity specification is very flexible, a semi-parametric approach could be an alternative. The dynamics within this panel survival model are modelled in a rudimentary way including only the previous lagged durations. A specification relying on the autoregressive conditional intensity model of Russell (1999) could also be a reasonable alternative. A time dependent factor capturing unobservable heterogeneity might help to improve the goodness-of-fit of the model. From an application point of view, a better categorization of large and small, experienced and inexperienced, patient and impatient as well as cautious and not-cautious investors would be desirable to obtain a better discrimination of behavioral biases across the investor groups. However, all of these tasks are left for further research on individual trading behavior and trading activity datasets.

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Notes

¹See for example Odean (1998a), Grinblatt and Keloharju (2001) Feng and Seasholes (2005), Locke and Onayev (2005), Dhar and Zhu (2006), O'Connell and Teo (2009).

²See Nolte and Voev (2011) for a deeper discussion and relaxation of this assumption.

³A key concept in prospect theory is that investors assess profits and losses relative to a reference point, which is usually considered to be a function of the assets' purchase prices. It is assumed that investors behave risk averse when their asset position has established a paper profit, and risk seeking when their position suffers from a paper loss.

⁴c.f. Feng and Seasholes (2005), Ivkovich et al. (2005), Locke and Mann (2005), Locke and Onayev (2005) and Shumway and Wu (2005).

⁵A more elaborate description of the trading mechanism on OANDA FXTrade and an analysis of the relationship between order flow and price changes can be found in Nolte and Nolte (2011).

⁶The full set of statistics and figures for all 20 investor groups are available from the author upon request.

⁷In particular, when being compared to the residual series of standard baseline intensity specifications without FFF augmentation that were used as benchmark specifications in a previous version of the paper.

⁸See also the descriptive statistics in Table 4.

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Appendix

Descriptive Statistics

EUR/USD	durations in hours				Profit/ Loss	absolute Profit/Loss	volume	dummy variable	mean
	roundtrip	inactivity	loss	profit					
Group 1: (0%-5%), # investors 138									
observations	4348.00	4283.00	1723.00	2518.00	4348.00	4348.00	4348.00	complex trading strategy	0.50
mean	30.88	58.97	33.22	30.49	-0.05	0.64	144.18		
std. deviation	110.89	216.12	127.47	100.40	2.53	2.45	244.20	close special limit order	0.21
minimum	0.00	0.00	0.00	0.00	-67.32	0.00	1.00		
5% Quantile	0.02	0.00	0.01	0.03	-1.11	0.00	1.00	successful investor	0.41
25% Quantile	0.25	0.05	0.17	0.34	-0.07	0.01	17.00		
50% Quantile	1.99	2.73	1.80	2.31	0.01	0.10	81.00	successful investor	0.41
75% Quantile	17.27	26.31	15.87	19.27	0.13	0.42	170.00		
95% Quantile	137.84	237.67	146.65	136.86	1.35	2.19	500.00	successful investor	0.41
maximum	2545.21	3722.03	2545.21	1989.13	27.43	67.32	4800.00		
Group 2: (5%-10%), # investors 142									
observations	6374.00	6297.00	2566.00	3604.00	6374.00	6374.00	6374.00	complex trading strategy	0.50
mean	22.92	42.51	24.79	22.63	-0.09	0.96	299.12		
std. deviation	121.53	193.55	158.93	90.03	4.37	4.26	409.72	close special limit order	0.20
minimum	0.00	0.00	0.00	0.00	-235.76	0.00	1.00		
5% Quantile	0.02	0.00	0.01	0.02	-1.92	0.00	5.00	successful investor	0.40
25% Quantile	0.15	0.06	0.12	0.21	-0.12	0.03	60.00		
50% Quantile	0.87	1.03	0.62	1.25	0.01	0.18	180.00	successful investor	0.40
75% Quantile	6.51	17.08	3.74	10.10	0.23	0.72	420.00		
95% Quantile	96.18	166.01	92.71	101.90	2.04	3.80	1000.00	successful investor	0.40
maximum	4233.81	4266.44	4233.81	1980.54	51.91	235.76	7000.00		
Group 3: (10%-15%), # investors 147									
observations	6318.00	6234.00	2611.00	3545.00	6318.00	6318.00	6318.00	complex trading strategy	0.44
mean	21.42	41.22	24.98	19.69	-0.30	2.46	713.61		
std. deviation	116.04	189.90	137.88	99.85	19.27	19.11	889.89	close special limit order	0.19
minimum	0.00	0.00	0.00	0.00	-686.46	0.00	1.00		
5% Quantile	0.02	0.01	0.02	0.02	-3.78	0.01	40.00	successful investor	0.34
25% Quantile	0.13	0.04	0.11	0.16	-0.39	0.12	200.00		
50% Quantile	0.74	0.89	0.60	0.91	0.04	0.47	500.00	successful investor	0.34
75% Quantile	6.08	15.49	4.51	8.41	0.52	1.60	1000.00		
95% Quantile	86.10	155.84	98.46	82.28	4.16	7.80	2000.00	successful investor	0.34
maximum	3389.15	4858.68	3389.15	2907.46	974.18	974.18	23000.00		
Group 4: (15%-20%), # investors 137									
observations	8308.00	8235.00	3320.00	4725.00	8308.00	8308.00	8308.00	complex trading strategy	0.48
mean	16.11	33.11	20.67	13.74	-0.48	2.57	933.77		
std. deviation	93.06	163.21	125.28	64.56	14.44	14.22	1083.03	close special limit order	0.18
minimum	0.00	0.00	0.00	0.00	-839.95	0.00	1.00		
5% Quantile	0.02	0.00	0.02	0.03	-5.43	0.01	25.00	successful investor	0.41
25% Quantile	0.14	0.03	0.13	0.16	-0.48	0.13	211.00		
50% Quantile	0.64	0.52	0.63	0.71	0.04	0.50	575.00	successful investor	0.41
75% Quantile	4.06	10.31	3.57	4.94	0.51	1.80	1032.00		
95% Quantile	67.25	130.52	72.36	64.56	4.40	8.98	3000.00	successful investor	0.41
maximum	4056.95	4568.80	4056.95	2517.26	634.25	839.95	16000.00		
Group 5: (20%-25%), # investors 142									
observations	9649.00	9563.00	3635.00	5666.00	9649.00	9649.00	9649.00	complex trading strategy	0.44
mean	16.29	27.46	21.63	13.75	-0.95	4.00	1295.80		
std. deviation	89.11	133.62	104.48	80.50	21.20	20.84	1813.66	close special limit order	0.19
minimum	0.00	0.00	0.00	0.00	-1169.60	0.00	1.00		
5% Quantile	0.02	0.00	0.02	0.03	-7.83	0.01	40.00	successful investor	0.42
25% Quantile	0.16	0.04	0.16	0.17	-0.52	0.12	200.00		
50% Quantile	0.69	0.78	0.72	0.74	0.06	0.60	900.00	successful investor	0.42
75% Quantile	4.54	12.60	5.00	4.88	0.70	2.40	1698.50		
95% Quantile	70.44	104.34	96.22	64.83	6.76	14.60	4094.15	successful investor	0.42
maximum	4031.26	4528.48	2595.92	4031.26	260.42	1169.60	50000.00		

Table 4: Descriptive statistics of the roundtrip, inactivity durations and explanatory variables for the EUR/USD currency pair. The sample ranges from 1st October 2003 00:00:00 (EST) to 14th May 2004 23:59:59 (EST), which are 227 full days (5448 hours).

EUR/USD	durations in hours				Profit/ Loss	absolute Profit/Loss	volume	dummy variable	mean
	roundtrip	inactivity	loss	profit					
Group 6: (25%-30%), # investors 146									
observations	10669.00	10579.00	4330.00	5855.00	10669.00	10669.00	10669.00	complex trading strategy	0.42
mean	13.98	26.08	17.24	12.60	-1.27	5.21	1821.73	close special limit order	
std. deviation	71.33	115.00	81.24	66.08	23.49	22.94	2557.32		
minimum	0.00	0.00	0.00	0.00	-765.85	0.00	1.00	successful investor	0.18
5% Quantile	0.02	0.00	0.01	0.02	-11.50	0.00	25.00		
25% Quantile	0.13	0.05	0.11	0.14	-0.78	0.16	400.00		
50% Quantile	0.54	0.69	0.50	0.64	0.03	0.80	1000.00	successful investor	0.39
75% Quantile	3.28	10.54	3.12	3.87	0.80	3.00	2191.50		
95% Quantile	64.77	117.77	75.48	59.00	9.00	19.83	6000.00		
maximum	3067.68	3771.70	1850.84	3067.68	549.30	765.85	100000.00		
Group 7: (30%-35%), # investors 141									
observations	13067.00	12986.00	5318.00	7284.00	13067.00	13067.00	13067.00	complex trading strategy	0.39
mean	11.14	19.89	12.20	10.83	-0.64	5.70	2076.74	close special limit order	
std. deviation	63.69	106.03	77.85	52.88	35.23	34.78	2841.86		
minimum	0.00	0.00	0.00	0.00	-1517.50	0.00	1.00	successful investor	0.19
5% Quantile	0.02	0.00	0.01	0.02	-10.00	0.00	90.00		
25% Quantile	0.11	0.04	0.10	0.13	-0.92	0.25	500.00		
50% Quantile	0.48	0.47	0.44	0.58	0.06	1.00	1000.00	successful investor	0.39
75% Quantile	2.66	7.85	2.32	3.16	1.00	3.20	2500.00		
95% Quantile	46.44	81.66	44.65	48.76	8.80	18.28	7000.00		
maximum	2325.87	4201.72	2325.87	2035.47	1771.67	1771.67	50000.00		
Group 8: (35%-40%), # investors 141									
observations	13083.00	12994.00	5392.00	7116.00	13083.00	13083.00	13083.00	complex trading strategy	0.42
mean	9.97	20.99	10.27	10.40	-1.27	8.36	3001.59	close special limit order	
std. deviation	64.88	106.93	67.87	65.07	54.31	53.68	4166.86		
minimum	0.00	0.00	0.00	0.00	-3798.30	0.00	1.00	successful investor	0.16
5% Quantile	0.02	0.00	0.01	0.02	-15.00	0.00	40.00		
25% Quantile	0.11	0.04	0.10	0.12	-0.90	0.20	500.00		
50% Quantile	0.42	0.54	0.38	0.47	0.04	1.00	1240.00	successful investor	0.38
75% Quantile	2.06	8.73	1.75	2.62	1.20	4.40	4000.00		
95% Quantile	34.97	90.83	34.41	40.28	13.80	29.05	10000.00		
maximum	3166.38	4024.91	2011.06	3166.38	1877.60	3798.30	60000.00		
Group 9: (40%-45%), # investors 149									
observations	14464.00	14369.00	5343.00	8614.00	14464.00	14464.00	14464.00	complex trading strategy	0.43
mean	9.34	19.37	11.05	8.73	-1.44	9.14	3806.51	close special limit order	
std. deviation	51.15	103.64	63.36	43.54	50.85	50.05	5673.49		
minimum	0.00	0.00	0.00	0.00	-2397.20	0.00	1.00	successful investor	0.18
5% Quantile	0.02	0.01	0.02	0.02	-17.39	0.00	50.00		
25% Quantile	0.13	0.05	0.13	0.13	-0.84	0.30	575.00		
50% Quantile	0.50	0.42	0.54	0.51	0.12	1.30	2000.00	successful investor	0.44
75% Quantile	2.52	6.03	2.74	2.62	1.65	5.60	5000.00		
95% Quantile	38.39	85.45	44.83	37.10	17.00	32.00	12000.00		
maximum	1677.39	5025.12	1677.39	1581.63	691.00	2397.20	120000.00		
Group 10: (45%-50%), # investors 146									
observations	14678.00	14596.00	5795.00	8253.00	14678.00	14678.00	14678.00	complex trading strategy	0.36
mean	10.77	19.54	13.20	9.66	-1.79	11.75	4630.25	close special limit order	
std. deviation	65.76	105.01	85.06	50.79	56.35	55.14	6923.37		
minimum	0.00	0.00	0.00	0.00	-2598.16	0.00	1.00	successful investor	0.18
5% Quantile	0.02	0.01	0.02	0.02	-22.82	0.00	71.00		
25% Quantile	0.13	0.05	0.12	0.13	-1.04	0.32	798.50		
50% Quantile	0.52	0.50	0.54	0.55	0.12	1.60	2280.00	successful investor	0.40
75% Quantile	2.88	7.23	2.95	3.08	2.00	7.00	5148.50		
95% Quantile	45.54	77.61	46.27	47.39	21.00	43.00	19010.00		
maximum	2225.20	4650.45	2225.20	2145.47	1040.64	2598.16	125027.00		

Table 4 (cont'd): Descriptive statistics of the roundtrip, inactivity durations and explanatory variables for the EUR/USD currency pair. The sample ranges from 1st October 2003 00:00:00 (EST) to 14th May 2004 23:59:59 (EST), which are 227 full days (5448 hours).

EUR/USD	durations in hours				Profit/ Loss	absolute Profit/Loss	volume	dummy variable	mean
	roundtrip	inactivity	loss	profit					
Group 11: (50%-55%), # investors 146									
observations	14824.00	14749.00	5881.00	8349.00	14824.00	14824.00	14824.00	complex trading strategy	0.40
mean	10.72	20.51	12.58	10.06	-2.38	15.20	6017.88		
std. deviation	62.17	102.80	73.33	55.35	80.69	79.28	8732.65		0.20
minimum	0.00	0.00	0.00	0.00	-3027.00	0.00	1.00	close	
5% Quantile	0.02	0.01	0.01	0.02	-28.00	0.00	65.00	special limit order	0.40
25% Quantile	0.12	0.05	0.11	0.14	-1.90	0.50	1000.00		
50% Quantile	0.49	0.46	0.44	0.57	0.11	2.30	3000.00		0.40
75% Quantile	2.70	6.80	2.49	3.12	2.73	9.00	8997.00		
95% Quantile	43.11	90.84	47.76	42.23	26.29	55.00	20000.00	successful investor	0.40
maximum	2185.84	3369.36	2185.84	1858.51	2760.20	3027.00	310000.00		
Group 12: (55%-60%), # investors 144									
observations	14845.00	14759.00	5706.00	8453.00	14845.00	14845.00	14845.00	complex trading strategy	0.37
mean	10.84	20.77	14.47	9.10	-3.85	18.92	7169.12		
std. deviation	65.68	97.34	80.90	55.89	98.18	96.42	10436.24		0.22
minimum	0.00	0.00	0.00	0.00	-3421.40	0.00	1.00	close	
5% Quantile	0.02	0.01	0.02	0.03	-33.00	0.00	180.00	special limit order	0.41
25% Quantile	0.12	0.07	0.12	0.13	-2.00	0.50	1000.00		
50% Quantile	0.43	0.50	0.44	0.45	0.20	2.40	3250.00		0.41
75% Quantile	2.18	7.51	2.29	2.31	2.88	9.79	10000.00		
95% Quantile	46.26	91.57	64.50	38.82	30.18	66.50	27000.00	successful investor	0.41
maximum	2373.72	3034.45	2137.58	2373.72	2552.80	3421.40	300000.00		
Group 13: (60%-65%), # investors 147									
observations	18939.00	18842.00	8053.00	10103.00	18939.00	18939.00	18939.00	complex trading strategy	0.31
mean	8.42	15.26	9.57	8.05	-2.99	18.95	8350.72		
std. deviation	54.63	78.73	66.33	45.59	123.18	121.75	14026.45		0.18
minimum	0.00	0.00	0.00	0.00	-5893.00	0.00	1.00	close	
5% Quantile	0.02	0.00	0.02	0.02	-35.20	0.00	100.00	special limit order	0.36
25% Quantile	0.11	0.04	0.10	0.12	-2.80	0.60	1047.75		
50% Quantile	0.41	0.28	0.40	0.45	0.10	3.00	5000.00		0.36
75% Quantile	1.94	3.99	1.83	2.24	3.30	11.00	10000.00		
95% Quantile	27.28	67.58	31.05	28.20	29.00	62.00	30000.00	successful investor	0.36
maximum	2661.71	2342.16	2661.71	1914.52	7395.00	7395.00	450000.00		
Group 14: (65%-70%), # investors 147									
observations	18462.00	18366.00	7809.00	9782.00	18462.00	18462.00	18462.00	complex trading strategy	0.37
mean	7.41	16.13	7.38	7.95	-3.32	26.72	11921.28		
std. deviation	45.15	78.60	44.65	47.41	159.78	157.56	17266.69		0.17
minimum	0.00	0.00	0.00	0.00	-9412.50	0.00	1.00	close	
5% Quantile	0.02	0.01	0.01	0.03	-45.00	0.00	360.20	special limit order	0.36
25% Quantile	0.10	0.05	0.09	0.12	-4.20	1.09	2700.00		
50% Quantile	0.34	0.37	0.30	0.41	0.20	4.72	7000.00		0.36
75% Quantile	1.53	5.30	1.35	1.88	5.00	15.00	14050.00		
95% Quantile	25.65	71.58	26.31	27.85	45.00	92.00	40021.20	over-confident investor	0.36
maximum	1895.68	3005.62	1895.68	1831.42	9162.00	9412.50	400000.00		
Group 15: (70%-75%), # investors 146									
observations	21653.00	21574.00	8818.00	11878.00	21653.00	21653.00	21653.00	complex trading strategy	0.39
mean	7.23	13.78	8.23	6.96	-5.15	28.35	12325.48		
std. deviation	47.60	73.51	58.54	39.74	234.58	232.91	19915.76		0.18
minimum	0.00	0.00	0.00	0.00	-24175.85	0.00	1.00	close	
5% Quantile	0.02	0.00	0.01	0.02	-50.57	0.00	100.00	special limit order	0.38
25% Quantile	0.09	0.03	0.08	0.11	-3.00	0.66	1200.00		
50% Quantile	0.35	0.23	0.33	0.40	0.14	4.00	6000.00		0.38
75% Quantile	1.56	3.51	1.45	1.81	4.50	16.20	15000.00		
95% Quantile	23.36	65.74	23.37	24.75	46.00	95.00	50000.00	successful investor	0.38
maximum	1869.31	2396.31	1869.31	1544.82	4196.34	24175.85	1000000.00		

Table 4 (cont'd): Descriptive statistics of the roundtrip, inactivity durations and explanatory variables for the EUR/USD currency pair. The sample ranges from 1st October 2003 00:00:00 (EST) to 14th May 2004 23:59:59 (EST), which are 227 full days (5448 hours).

EUR/USD	durations in hours				Profit/ Loss	absolute Profit/Loss	volume	dummy variable	mean
	roundtrip	inactivity	loss	profit					
Group 16: (75%-80%), # investors 146									
observations	23314.00	23224.00	9502.00	12798.00	23314.00	23314.00	23314.00	complex trading strategy	0.29
mean	6.93	13.93	7.54	6.89	-8.25	37.19	17838.46	close special limit order	0.17
std. deviation	39.39	73.25	45.28	35.99	175.54	171.75	26604.71		
minimum	0.00	0.00	0.00	0.00	-6623.00	0.00	1.00		
5% Quantile	0.02	0.00	0.01	0.02	-79.82	0.00	432.70		
25% Quantile	0.11	0.03	0.10	0.12	-6.00	1.54	3144.00	successful investor	0.38
50% Quantile	0.38	0.23	0.38	0.41	0.35	6.50	10000.00		
75% Quantile	1.88	3.14	1.79	2.12	7.10	22.66	20000.00		
95% Quantile	25.49	66.99	26.93	26.15	63.00	137.03	60000.00		
maximum	1682.17	2374.48	1682.17	1488.00	2421.40	6623.00	500000.00		
Group 17: (80%-85%), # investors 147									
observations	23522.00	23412.00	8917.00	13508.00	23522.00	23522.00	23522.00	complex trading strategy	0.39
mean	6.39	13.85	7.74	5.93	-7.20	46.77	24291.91	close special limit order	0.24
std. deviation	33.96	74.53	43.38	27.58	196.01	190.49	32220.84		
minimum	0.00	0.00	0.00	0.00	-10151.60	0.00	1.00		
5% Quantile	0.02	0.01	0.01	0.02	-100.00	0.01	1000.00		
25% Quantile	0.11	0.04	0.10	0.12	-7.60	2.04	5000.00	successful investor	0.42
50% Quantile	0.39	0.28	0.38	0.43	0.90	10.00	13500.00		
75% Quantile	1.78	4.29	1.73	1.96	11.00	33.00	30000.00		
95% Quantile	23.78	68.00	27.99	23.86	85.00	179.87	100000.00		
maximum	1534.95	3447.64	1534.95	886.31	3590.00	10151.60	1040400.00		
Group 18: (85%-90%), # investors 145									
observations	20982.00	20884.00	8117.00	11823.00	20982.00	20982.00	20982.00	complex trading strategy	0.36
mean	6.82	14.48	8.42	6.21	-15.48	85.08	40069.26	close special limit order	0.19
std. deviation	45.85	72.50	58.38	37.18	543.92	537.45	52697.31		
minimum	0.00	0.00	0.00	0.00	-45935.70	0.00	1.00		
5% Quantile	0.02	0.01	0.02	0.02	-166.50	0.00	1161.10		
25% Quantile	0.10	0.05	0.10	0.10	-16.69	4.02	10000.00	successful investor	0.40
50% Quantile	0.36	0.41	0.38	0.37	1.30	18.00	23329.00		
75% Quantile	1.68	5.36	1.70	1.81	18.18	56.12	50000.00		
95% Quantile	23.23	69.43	26.45	22.56	140.00	279.00	125000.00		
maximum	2232.12	3465.18	2232.12	1555.09	15169.50	45935.70	1200000.00		
Group 19: (90%-95%), # investors 146									
observations	23261.00	23180.00	8378.00	13910.00	23261.00	23261.00	23261.00	complex trading strategy	0.25
mean	7.41	12.91	9.04	6.82	-12.93	126.84	62529.85	close special limit order	0.26
std. deviation	41.42	72.74	49.41	37.28	575.81	561.81	85607.01		
minimum	0.00	0.00	0.00	0.00	-18630.00	0.00	1.00		
5% Quantile	0.02	0.01	0.02	0.02	-250.00	0.02	1000.00		
25% Quantile	0.11	0.04	0.11	0.12	-12.50	4.60	10000.00	successful investor	0.45
50% Quantile	0.40	0.29	0.42	0.41	2.05	20.00	30000.00		
75% Quantile	1.93	3.26	2.18	1.95	28.00	80.00	90462.00		
95% Quantile	27.68	62.76	35.91	26.72	240.00	478.39	200000.00		
maximum	1824.01	2480.13	1698.30	1824.01	34727.50	34727.50	1200000.00		
Group 20: (95%-100%), # investors 149									
observations	26325.00	26237.00	9172.00	15910.00	26325.00	26325.00	26325.00	complex trading strategy	0.26
mean	7.81	12.37	10.14	7.00	-53.40	880.21	366229.32	close special limit order	0.25
std. deviation	52.41	62.10	67.75	43.46	10674.13	10637.91	1047316.73		
minimum	0.00	0.00	0.00	0.00	-809400.00	0.00	1.00		
5% Quantile	0.02	0.01	0.01	0.02	-858.75	0.05	6000.00		
25% Quantile	0.09	0.04	0.10	0.10	-41.42	16.50	45000.00	successful investor	0.46
50% Quantile	0.35	0.30	0.41	0.34	10.00	76.62	100000.00		
75% Quantile	1.69	3.75	2.12	1.66	100.00	295.00	300000.00		
95% Quantile	26.86	63.28	41.78	24.37	1140.00	2350.00	1400000.00		
maximum	3284.07	3485.18	3284.07	1296.50	688400.00	809400.00	53000000.00		

Table 4 (cont'd): Descriptive statistics of the roundtrip, inactivity durations and explanatory variables for the EUR/USD currency pair. The sample ranges from 1st October 2003 00:00:00 (EST) to 14th May 2004 23:59:59 (EST), which are 227 full days (5448 hours).

Estimation Results

Par.	1		2		3		4		5	
	Est.	Std.	Est.	Std.	Est.	Std.	Est.	Std.	Est.	Std.
α^R	Roundtrip Intensity									
	Baseline Intensity									
	0.9541	0.0460	1.0078	0.0316	1.1047	0.0358	1.0614	0.0177	1.0782	0.0176
	$\sigma^{2,R}$	0.0557	0.1126	0.1874	0.0682	0.2908	0.0578	0.1542	0.0507	0.2285
	$\nu_{s,1}^R$	-0.1735	0.0458	-0.1321	0.0445	-0.1063	0.0525	0.0035	0.0512	-0.0708
	$\nu_{s,2}^R$	-0.0002	0.1312	0.1181	0.0412	0.0038	0.1794	0.0154	0.0344	0.0386
	$\nu_{s,3}^R$	-0.2197	0.0377	-0.1825	0.0311	-0.2034	0.0593	-0.2144	0.0410	-0.1589
	$\nu_{c,1}^R$	0.0364	0.0381	0.0466	0.0326	0.0664	0.0338	0.0510	0.0451	-0.0060
	$\nu_{c,2}^R$	0.1780	0.0324	0.1963	0.0308	0.1753	0.0392	0.2010	0.0461	0.2401
	$\nu_{c,3}^R$	0.0956	0.0417	0.0924	0.0316	0.1263	0.0704	0.0888	0.0312	0.1271
c^R	Explanatory Variables									
	2.6288	0.6426	2.3556	0.3248	1.2231	0.4646	1.0073	0.1035	1.3277	0.0831
	c_{sc}^R	-0.7485	0.3105	-0.2360	0.1141	-0.2728	0.3491	-0.5841	0.0527	-0.5697
	c_{sc}^{sl}	-0.1559	0.1996	-0.0541	0.1319	0.1935	0.1518	-0.0634	0.0611	-0.0073
	β_p^R	-3.3570	0.4153	-2.8717	0.3019	-2.7995	0.2447	-2.0900	0.1053	-2.5563
	β_{p*sl}^R	0.4062	0.1478	0.0180	0.0308	0.0204	0.2916	0.1654	0.0398	0.1173
	β_{p*ov}^R	0.0541	0.1032	-0.0262	0.0749	-0.1609	0.0874	-0.0512	0.0517	-0.0071
	β_l^R	2.3996	0.3372	1.4166	0.2659	1.6355	0.1908	0.9926	0.1043	1.4372
	β_{l*sl}^R	-0.4813	0.1359	-0.1168	0.0629	-0.2301	0.1515	-0.3119	0.0365	-0.2785
	β_{l*ov}^R	0.0389	0.0929	0.0847	0.0656	0.1225	0.0767	0.0361	0.0465	0.0341
	$\beta_{p^2}^R$	0.7994	0.1706	0.8554	0.1589	0.5869	0.1097	0.3530	0.0468	0.6003
	$\beta_{l^2}^R$	0.2993	0.1218	0.1099	0.1145	0.0225	0.0791	-0.1540	0.0554	0.0231
	$\beta_{p^3}^R$	-0.0808	0.0218	-0.1068	0.0232	-0.0556	0.0136	-0.0398	0.0060	-0.0720
	$\beta_{l^3}^R$	0.0162	0.0141	0.0150	0.0126	0.0003	0.0079	-0.0122	0.0082	0.0021
	β_{p*cx}^R	0.0657	0.0879	-0.0366	0.1064	0.0589	0.1047	0.0201	0.0513	0.0005
	β_{l*cx}^R	-0.1092	0.0742	-0.0713	0.0824	-0.1366	0.0856	-0.1529	0.0514	-0.0012
	$\beta_{dr(-1)}^R$	-0.2493	0.0344	-0.2536	0.0233	-0.2502	0.0236	-0.1860	0.0087	-0.2190
	$\beta_{di(-1)}^R$	-0.0840	0.0198	-0.0739	0.0111	-0.0735	0.0133	-0.0614	0.0070	-0.0437
	β_{vol}^R	0.4157	0.0819	0.3854	0.0634	0.6209	0.0805	0.6159	0.0219	0.5423
	c_{cx}^R	-1.5016	0.1833	-1.4318	0.1554	-1.5252	0.1815	-1.6460	0.0901	-1.1892
α^I	Inactivity Intensity									
	Baseline Intensity									
	0.4453	0.0279	0.5163	0.0397	0.5230	0.0359	0.5114	0.0084	0.4917	0.0085
	$\sigma^{2,I}$	-0.5886	0.2043	-0.2719	0.2074	-0.2368	0.1614	-0.2402	0.0546	-0.4379
	$\nu_{s,1}^I$	-0.1005	0.0589	-0.1276	0.0567	-0.1361	0.0482	-0.0536	0.0388	-0.1616
	$\nu_{s,2}^I$	-0.1673	0.0428	0.0147	0.0816	-0.0665	0.0418	-0.1114	0.0283	-0.0803
	$\nu_{s,3}^I$	-0.3750	0.0394	-0.4202	0.0410	-0.4099	0.0396	-0.4367	0.0345	-0.4858
	$\nu_{c,1}^I$	-0.0652	0.0315	-0.0644	0.0324	-0.0793	0.0263	-0.1105	0.0281	-0.0659
	$\nu_{c,2}^I$	0.2576	0.0433	0.3854	0.0374	0.3758	0.0412	0.3533	0.0293	0.3244
	$\nu_{c,3}^I$	0.1608	0.0346	0.1148	0.0360	0.0541	0.0374	0.0893	0.0252	0.0330
	Explanatory Variables									
	c^I	-0.0059	0.0375	0.2169	0.1691	0.2677	0.1069	0.3237	0.0573	0.3365
	c_{sc}^I	0.1988	0.0945	-0.0121	0.1733	0.0093	0.0747	0.1476	0.0354	0.0618
	$\beta_{dr(-1)}^I$	-0.1818	0.0262	-0.1778	0.0261	-0.2184	0.0306	-0.2044	0.0064	-0.1717
	$\beta_{di(-1)}^I$	-0.1371	0.0218	-0.1555	0.0228	-0.1462	0.0160	-0.1551	0.0051	-0.1544

Table 5: Estimation results for investor groups 1 to 5 for EUR/USD. Quasi-maximum likelihood standard errors reported. Subindexes: p – profit, l – loss, sl – special limit order, sc – successful, cx – complex trading strategy, $dr(-1)$ – lagged roundtrip duration, $di(-1)$ – lagged inactivity duration, vol – transaction volume, $*$ – respective interaction term. All other parameters are detailed in the main text.

Par.	6		7		8		9		10	
	Est.	Std.	Est.	Std.	Est.	Std.	Est.	Std.	Est.	Std.
α^R $\sigma^{2,R}$ $\nu_{s,1}^R$ $\nu_{s,2}^R$ $\nu_{s,3}^R$ $\nu_{c,1}^R$ $\nu_{c,2}^R$ $\nu_{c,3}^R$	Roundtrip Intensity									
	Baseline Intensity									
	0.9972	0.0354	1.0559	0.0167	1.0501	0.0241	1.1202	0.0150	1.0328	0.0141
	0.0841	0.0687	0.1103	0.0434	0.1114	0.0587	0.2108	0.0328	0.0839	0.0355
	-0.0904	0.0396	-0.0262	0.0317	-0.0965	0.0424	-0.1208	0.0356	-0.0372	0.0351
	0.0270	0.0321	-0.0084	0.0299	0.0481	0.0286	0.0202	0.0301	0.0065	0.0303
	-0.1561	0.0287	-0.1558	0.0280	-0.1815	0.0279	-0.2106	0.0300	-0.2505	0.0268
	0.0652	0.0308	0.0399	0.0311	0.0546	0.0325	0.0623	0.0228	0.0928	0.0235
	0.2673	0.0305	0.2118	0.0266	0.1770	0.0366	0.2141	0.0291	0.2155	0.0300
	0.1060	0.0280	0.0306	0.0279	0.1099	0.0295	0.0058	0.0267	-0.0156	0.0286
c^R c_{sl}^R c_{sc}^R β_p^R β_{p*sl}^R β_{p*ov}^R β_l^R β_{l*sl}^R β_{l*ov}^R β_p^2 $\beta_{l^2}^R$ $\beta_{l^3}^R$ $\beta_{p^3}^R$ β_{p*cx}^R β_{l*cx}^R $\beta_{dr(-1)}^R$ $\beta_{di(-1)}^R$ β_{vol}^R c_{cx}^R	Explanatory Variables									
	1.7223	0.4811	0.9290	0.0533	1.9464	0.4244	1.6318	0.0571	1.1997	0.0593
	-0.5079	0.1329	-0.3105	0.0278	-0.2726	0.1418	-0.3259	0.0247	-0.5697	0.0286
	-0.0558	0.1316	0.3303	0.0458	0.1045	0.0636	0.1127	0.0373	-0.0333	0.0466
	-1.7693	0.2289	-1.7124	0.0642	-1.4313	0.1860	-1.7335	0.0640	-1.7625	0.0565
	0.1740	0.0666	-0.0257	0.0282	-0.0669	0.0628	-0.0185	0.0270	0.0824	0.0267
	0.0300	0.0809	-0.2293	0.0453	-0.1161	0.0412	-0.0448	0.0341	0.0465	0.0361
	0.8816	0.1855	0.6402	0.0521	0.7167	0.1721	0.7886	0.0701	0.9082	0.0685
	-0.2537	0.0626	-0.1581	0.0273	-0.1176	0.0745	-0.0910	0.0236	-0.1893	0.0219
	0.0828	0.0577	0.2316	0.0413	0.0963	0.0379	0.1567	0.0338	-0.0062	0.0382
	0.2980	0.0876	0.3264	0.0301	0.2854	0.0782	0.3493	0.0388	0.3548	0.0310
	-0.0688	0.0811	-0.1858	0.0266	-0.0591	0.0707	-0.0589	0.0382	-0.0304	0.0377
	-0.0344	0.0114	-0.0351	0.0036	-0.0374	0.0102	-0.0463	0.0064	-0.0419	0.0050
	-0.0062	0.0094	-0.0190	0.0037	0.0004	0.0078	0.0001	0.0054	0.0021	0.0056
	0.0259	0.0630	-0.0522	0.0314	-0.0239	0.0443	0.1533	0.0345	-0.0958	0.0309
	0.0429	0.0560	-0.0030	0.0349	-0.0973	0.0491	-0.1567	0.0294	0.0572	0.0323
	-0.2328	0.0201	-0.2469	0.0058	-0.2352	0.0166	-0.2547	0.0067	-0.2451	0.0063
	-0.0543	0.0094	-0.0995	0.0046	-0.0857	0.0114	-0.0514	0.0039	-0.0622	0.0045
	0.3250	0.0943	0.4508	0.0104	0.2802	0.0584	0.4062	0.0077	0.3889	0.0089
	-1.1377	0.1373	-1.1857	0.0404	-1.2873	0.1014	-1.6842	0.0590	-1.0183	0.0389
α^I $\sigma^{2,I}$ $\nu_{s,1}^I$ $\nu_{s,2}^I$ $\nu_{s,3}^I$ $\nu_{c,1}^I$ $\nu_{c,2}^I$ $\nu_{c,3}^I$	Inactivity Intensity									
	Baseline Intensity									
	0.5334	0.0231	0.6470	0.0102	0.5504	0.0319	0.6032	0.0053	0.5949	0.0069
	-0.2904	0.1062	0.0161	0.0434	-0.2725	0.1509	-0.1332	0.0283	-0.1896	0.0387
	-0.1003	0.0544	-0.1223	0.0362	-0.1344	0.0544	-0.0538	0.0321	-0.0688	0.0290
	-0.0351	0.0373	-0.1396	0.0220	-0.0404	0.0407	-0.0647	0.0224	-0.0912	0.0222
	-0.3650	0.0337	-0.4836	0.0251	-0.4738	0.0393	-0.4093	0.0182	-0.4630	0.0259
	-0.1047	0.0240	-0.1072	0.0287	-0.0873	0.0286	-0.0365	0.0241	-0.0646	0.0228
	0.3578	0.0349	0.3658	0.0266	0.3792	0.0302	0.3876	0.0288	0.4580	0.0233
	0.1121	0.0318	0.0258	0.0181	0.0573	0.0393	0.0649	0.0206	-0.0061	0.0188
c^I c_{sc}^I $\beta_{dr(-1)}^I$ $\beta_{di(-1)}^I$	Explanatory Variables									
	0.4580	0.1021	0.6843	0.0591	0.3655	0.0974	0.5761	0.0449	0.5684	0.0417
	-0.1461	0.0612	-0.0272	0.0320	0.0123	0.0579	0.0011	0.0205	0.0013	0.0264
	-0.1559	0.0174	-0.2427	0.0049	-0.1992	0.0213	-0.2003	0.0049	-0.1787	0.0047
	-0.1679	0.0159	-0.2065	0.0049	-0.1656	0.0227	-0.1634	0.0031	-0.1787	0.0038

Table 5 (cont'd): Estimation results for investor groups 6 to 10 for EUR/USD. Quasi-maximum likelihood standard errors reported. Subindexes: p – profit, l – loss, sl –special limit order, sc – successfull, cx – complex trading strategy, $dr(-1)$ – lagged roundtrip duration, $di(-1)$ – lagged inactivity duration, vol – transaction volume, $*$ – respective interaction term. All other parameters are detailed in the main text.

Par.	11		12		13		14		15	
	Est.	Std.	Est.	Std.	Est.	Std.	Est.	Std.	Est.	Std.
α^R $\sigma^{2,R}$ $\nu_{s,1}^R$ $\nu_{s,2}^R$ $\nu_{s,3}^R$ $\nu_{c,1}^R$ $\nu_{c,2}^R$ $\nu_{c,3}^R$	Roundtrip Intensity									
	Baseline Intensity									
	1.0615	0.0122	1.0927	0.0310	1.1246	0.0137	1.1018	0.0138	1.0880	0.0092
	0.1075	0.0322	0.2001	0.0560	0.2360	0.0335	0.1573	0.0347	0.1170	0.0271
	-0.0750	0.0289	-0.0513	0.0337	0.0288	0.0339	-0.0562	0.0303	-0.0331	0.0319
	-0.0152	0.0306	0.0307	0.0288	-0.0095	0.0300	0.0224	0.0260	0.0444	0.0258
	-0.1854	0.0280	-0.2335	0.0260	-0.2426	0.0240	-0.2336	0.0268	-0.2224	0.0225
	0.0462	0.0260	0.1119	0.0288	0.0588	0.0285	0.1270	0.0305	0.0955	0.0272
	0.1815	0.0290	0.2719	0.0317	0.2595	0.0352	0.2593	0.0282	0.2121	0.0316
	0.0744	0.0254	0.1121	0.0266	0.0200	0.0275	0.0575	0.0231	0.0147	0.0261
c^R c_{sl}^R c_{sc}^R β_p^R β_{p*sl}^R β_{p*ov}^R β_l^R β_{l*sl}^R β_{l*ov}^R $\beta_{p^2}^R$ $\beta_{l^2}^R$ $\beta_{p^3}^R$ $\beta_{l^3}^R$ β_{p*cx}^R β_{l*cx}^R $\beta_{dr(-1)}^R$ $\beta_{di(-1)}^R$ β_{vol}^R c_{cx}^R	Explanatory Variables									
	1.6324	0.0559	1.7587	0.4326	2.1036	0.0685	2.6214	0.0714	2.3337	0.0653
	-0.6105	0.0251	-0.3817	0.1280	-0.7768	0.0346	-0.5720	0.0285	-0.4129	0.0222
	0.1410	0.0387	0.0996	0.0722	0.0827	0.0383	-0.0035	0.0455	0.0108	0.0300
	-1.3000	0.0585	-1.5209	0.1948	-1.2849	0.0485	-0.9109	0.0468	-1.1166	0.0540
	0.0849	0.0249	-0.0321	0.0693	0.1964	0.0269	0.1132	0.0201	0.0097	0.0229
	-0.1541	0.0344	-0.0906	0.0474	-0.0015	0.0306	0.0208	0.0294	-0.0151	0.0248
	0.4675	0.0527	0.6431	0.1888	0.5831	0.0514	0.2378	0.0502	0.4741	0.0470
	-0.2347	0.0229	-0.1802	0.0651	-0.2845	0.0279	-0.2707	0.0199	-0.0840	0.0219
	0.1833	0.0306	0.1459	0.0413	0.1121	0.0318	0.0504	0.0339	0.0923	0.0243
	0.2230	0.0359	0.3313	0.0881	0.1577	0.0253	0.0659	0.0257	0.1866	0.0317
	-0.1567	0.0298	-0.0818	0.0940	-0.1102	0.0250	-0.2405	0.0276	-0.0644	0.0215
	-0.0297	0.0053	-0.0415	0.0113	-0.0197	0.0036	-0.0127	0.0039	-0.0262	0.0047
	-0.0111	0.0040	-0.0020	0.0110	-0.0041	0.0033	-0.0200	0.0040	0.0005	0.0025
	-0.0191	0.0317	-0.0961	0.0729	0.0062	0.0282	-0.0517	0.0224	-0.0623	0.0224
	0.0132	0.0337	0.0689	0.0749	-0.0844	0.0287	0.0130	0.0268	0.0272	0.0270
	-0.2410	0.0052	-0.2107	0.0148	-0.2752	0.0045	-0.2642	0.0051	-0.3078	0.0050
	-0.1010	0.0041	-0.0818	0.0090	-0.0630	0.0034	-0.0876	0.0032	-0.0582	0.0031
	0.2997	0.0039	0.2898	0.0567	0.2809	0.0040	0.1437	0.0030	0.1883	0.0047
	-1.3427	0.0538	-1.2024	0.1578	-1.5603	0.0499	-1.1052	0.0308	-1.3336	0.0297
α^I $\sigma^{2,I}$ $\nu_{s,1}^I$ $\nu_{s,2}^I$ $\nu_{s,3}^I$ $\nu_{c,1}^I$ $\nu_{c,2}^I$ $\nu_{c,3}^I$	Inactivity Intensity									
	Baseline Intensity									
	0.6427	0.0067	0.6623	0.0244	0.6582	0.0074	0.6798	0.0060	0.6441	0.0068
	0.0030	0.0293	0.0200	0.0308	0.0483	0.0300	0.0742	0.0250	0.0689	0.0287
	-0.0053	0.0265	-0.1695	0.0379	-0.0855	0.0257	-0.1341	0.0267	-0.1854	0.0254
	-0.1212	0.0226	-0.0404	0.0291	-0.1306	0.0195	-0.0652	0.0192	-0.0232	0.0188
	-0.4246	0.0226	-0.4040	0.0280	-0.4380	0.0214	-0.4544	0.0177	-0.5296	0.0200
	-0.0452	0.0204	-0.0326	0.0232	-0.0720	0.0224	-0.0982	0.0249	-0.1016	0.0195
	0.3873	0.0221	0.4092	0.0347	0.3919	0.0232	0.5056	0.0211	0.4370	0.0234
	0.0785	0.0194	0.0919	0.0289	0.0203	0.0190	0.0124	0.0171	-0.0439	0.0165
c^I c_{sc}^I $\beta_{dr(-1)}^I$ $\beta_{di(-1)}^I$	Explanatory Variables									
	0.6284	0.0412	0.6489	0.0936	0.9503	0.0519	0.7517	0.0433	1.0667	0.0505
	-0.1059	0.0207	-0.0385	0.0437	-0.0017	0.0167	-0.0550	0.0122	-0.1267	0.0146
	-0.2411	0.0050	-0.2223	0.0161	-0.1899	0.0043	-0.2408	0.0043	-0.1735	0.0035
	-0.1692	0.0040	-0.1599	0.0125	-0.2063	0.0036	-0.1738	0.0030	-0.2209	0.0038

Table 5 (cont'd): Estimation results for investor groups 11 to 15 for EUR/USD. Quasi-maximum likelihood standard errors reported. Subindexes: p – profit, l – loss, sl –special limit order, sc – successfull, cx – complex trading strategy, $dr(-1)$ – lagged roundtrip duration, $di(-1)$ – lagged inactivity duration, vol – transaction volume, $*$ – respective interaction term. All other parameters are detailed in the main text.

Par.	16		17		18		19		20	
	Est.	Std.	Est.	Std.	Est.	Std.	Est.	Std.	Est.	Std.
α^R $\sigma^{2,R}$ $\nu_{s,1}^R$ $\nu_{s,2}^R$ $\nu_{s,3}^R$ $\nu_{c,1}^R$ $\nu_{c,2}^R$ $\nu_{c,3}^R$	Roundtrip Intensity									
	Baseline Intensity									
	1.0871	0.0086	1.1260	0.0138	1.1662	0.0126	1.1294	0.0094	1.0980	0.0099
	0.0887	0.0245	0.1818	0.0313	0.2351	0.0297	0.1979	0.0213	0.2292	0.0231
	-0.0806	0.0274	-0.0157	0.0292	-0.0832	0.0329	-0.0981	0.0290	-0.0510	0.0287
	0.0263	0.0222	-0.0203	0.0256	-0.0273	0.0280	0.0587	0.0199	0.0065	0.0240
	-0.2601	0.0196	-0.1915	0.0203	-0.2058	0.0233	-0.2773	0.0215	-0.2265	0.0243
	0.0838	0.0251	0.0913	0.0228	0.0773	0.0275	0.0978	0.0204	0.0886	0.0200
	0.2223	0.0272	0.2009	0.0227	0.2118	0.0273	0.2154	0.0237	0.2547	0.0274
	0.0854	0.0231	0.0262	0.0190	-0.0106	0.0198	0.0705	0.0212	0.0558	0.0182
c^R c_{sl}^R c_{sc}^R β_p^R β_{p*sl}^R β_{p*ov}^R β_l^R β_{l*sl}^R β_{l*ov}^R β_p^R $\beta_{l^2}^R$ $\beta_{l^3}^R$ $\beta_{p^3}^R$ β_{p*cx}^R β_{l*cx}^R $\beta_{dr(-1)}^R$ $\beta_{di(-1)}^R$ β_{vol}^R c_{cx}^R	Explanatory Variables									
	1.9495	0.0567	1.8576	0.0595	2.0001	0.0630	1.9345	0.0486	-0.1432	0.0412
	-0.7639	0.0240	-0.4868	0.0155	-0.5107	0.0182	-0.5809	0.0258	-0.4074	0.0155
	-0.0264	0.0342	0.2175	0.0317	0.1357	0.0328	0.2760	0.0254	0.0191	0.0186
	-1.2518	0.0363	-1.5617	0.0516	-1.2907	0.0543	-1.1904	0.0384	-1.8799	0.0392
	0.2481	0.0207	0.1174	0.0170	0.1241	0.0215	0.0803	0.0207	-0.2098	0.0226
	0.0147	0.0256	-0.0728	0.0223	-0.0653	0.0287	-0.1604	0.0214	-0.0068	0.0220
	0.5688	0.0411	0.7747	0.0449	0.6587	0.0470	0.6672	0.0494	1.3744	0.0404
	-0.3078	0.0161	-0.1669	0.0138	-0.1753	0.0192	-0.1676	0.0169	-0.1132	0.0204
	0.0841	0.0254	0.1443	0.0242	0.1229	0.0276	0.2571	0.0251	0.1060	0.0289
	0.2554	0.0244	0.2947	0.0300	0.1688	0.0318	0.2217	0.0218	0.4546	0.0188
	-0.0431	0.0225	-0.0201	0.0240	-0.0521	0.0243	0.0186	0.0265	0.1490	0.0210
	-0.0384	0.0040	-0.0394	0.0053	-0.0232	0.0052	-0.0324	0.0034	-0.0481	0.0024
	0.0025	0.0033	0.0046	0.0035	0.0018	0.0034	0.0087	0.0038	0.0125	0.0029
	-0.0338	0.0259	-0.0193	0.0197	0.0329	0.0300	-0.0982	0.0199	-0.0446	0.0206
	0.0274	0.0250	0.0574	0.0198	0.0245	0.0246	0.1041	0.0184	0.0021	0.0227
	-0.3345	0.0041	-0.2372	0.0051	-0.2789	0.0050	-0.2308	0.0040	-0.2219	0.0037
	-0.0481	0.0024	-0.0861	0.0036	-0.1037	0.0038	-0.0876	0.0029	-0.0905	0.0024
	0.2040	0.0038	0.2990	0.0066	0.2645	0.0046	0.2319	0.0040	0.4010	0.0054
	-1.2162	0.0495	-1.2305	0.0357	-1.4532	0.0453	-1.2298	0.0296	-1.5042	0.0267
α^I $\sigma^{2,I}$ $\nu_{s,1}^I$ $\nu_{s,2}^I$ $\nu_{s,3}^I$ $\nu_{c,1}^I$ $\nu_{c,2}^I$ $\nu_{c,3}^I$	Inactivity Intensity									
	Baseline Intensity									
	0.6316	0.0071	0.6682	0.0053	0.6392	0.0064	0.7061	0.0063	0.6737	0.0052
	0.0162	0.0315	0.0608	0.0223	-0.0812	0.0306	0.0932	0.0235	-0.0090	0.0224
	-0.1198	0.0254	-0.0726	0.0254	-0.1575	0.0274	-0.0279	0.0254	-0.1010	0.0197
	-0.0834	0.0163	-0.1457	0.0186	-0.0072	0.0144	-0.1596	0.0177	-0.1473	0.0191
	-0.4569	0.0211	-0.4984	0.0177	-0.4530	0.0210	-0.4349	0.0203	-0.4289	0.0149
	-0.0528	0.0188	-0.0803	0.0193	-0.1299	0.0159	-0.0772	0.0196	-0.0533	0.0180
	0.4609	0.0171	0.4661	0.0206	0.4049	0.0213	0.4762	0.0197	0.3502	0.0193
	-0.0043	0.0170	0.0451	0.0159	0.0802	0.0135	-0.0162	0.0159	0.0334	0.0146
	Explanatory Variables									
	0.8642	0.0467	0.8225	0.0416	0.7242	0.0362	1.0549	0.0392	1.0136	0.0433
	-0.0150	0.0152	-0.0525	0.0119	-0.0155	0.0149	0.0067	0.0120	-0.0059	0.0113
	-0.1723	0.0031	-0.2349	0.0038	-0.1995	0.0048	-0.1968	0.0042	-0.1705	0.0025
	-0.2073	0.0037	-0.1812	0.0022	-0.1845	0.0032	-0.2100	0.0030	-0.2110	0.0028
c^I c_{sc}^I $\beta_{dr(-1)}^I$ $\beta_{di(-1)}^I$										

Table 5 (cont'd): Estimation results for investor groups 16 to 20 for EUR/USD. Quasi-maximum likelihood standard errors reported. Subindexes: p – profit, l – loss, sl –special limit order, sc – successfull, cx – complex trading strategy, $dr(-1)$ – lagged roundtrip duration, $di(-1)$ – lagged inactivity duration, vol – transaction volume, $*$ – respective interaction term. All other parameters are detailed in the main text.